

Dual transcranial electromagnetic stimulation of the precuneus-hippocampus network boosts human long-term memory

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eLife Assessment

This work presents potentially **important** findings suggesting that a combination of transcranial stimulation approaches applied for a short period could improve memory performance. However, the evidence supporting the conclusions is currently **incomplete**. In particular, the claims relating to the specific neural mechanisms and anatomical sites of action underlying effects were viewed as overstated in the current version. The results potentially have implications for non-invasive enhancement of cognitive functions.

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Abstract

Non-invasive brain stimulation techniques have the potential to improve memory functions. However, the results so far have been relatively modest and time-consuming. Here, we implemented a novel 3-minute combination of personalized repetitive transcranial magnetic stimulation (intermittent theta burst-iTBS) coupled with simultaneous application of gamma transcranial alternating current stimulation (ytACS) over the precuneus, a brain area connected with the hippocampus, to modulate long term memory in healthy subjects. Only dual electromagnetic stimulation of the precuneus produced a consistent increase in long-term associative memory as compared to iTBS alone and sham conditions in a sample of healthy volunteers. The effects were replicated in another independent sample, in which the increased associative memory was retained for up to one week. Moreover, dual stimulation increased gamma oscillations and precuneus-hippocampus functional connectivity through the white matter tracts linking the precuneus with the temporal lobe. These findings show that dual stimulation may lead neuronal assemblies in a state favorable to enhance long-term plasticity and identify the precuneus as a key brain area involved in memory formation. Personalized dual electromagnetic stimulation of the precuneus-hippocampus network may

represent a new powerful approach for enhancing memory functions in several healthy and clinical conditions.

Introduction

As the world's population ages, the prevalence of age-related memory deficits is growing. This phenomenon often signals the onset of more severe cognitive decline which has a strong impact on society. In light of this escalating challenge, there is an urgent need for innovative strategies aimed at enhancing cognitive functions and potentially mitigating cognitive decline.

Non-invasive brain stimulation (NIBS) techniques have been largely used to enhance human cognition in the last two decades (Antal et al., 2022 [↗](#)). Encouraging results have been reported in both healthy and pathological conditions including depression, stroke, and Alzheimer's disease (Blumberger et al., 2018 [↗](#); Koch et al., 2022 [↗](#); Lefaucheur et al., 2020 [↗](#)).

Repetitive transcranial magnetic stimulation (rTMS) and transcranial alternating current stimulation (tACS) are two forms of NIBS widely used to enhance memory performances (Grover et al., 2022 [↗](#); Koch et al., 2018 [↗](#); Wang et al., 2014 [↗](#)). rTMS, based on the principle of Faraday, induces depolarization of cortical neuronal assemblies and leads to after-effects that have been linked to changes in synaptic plasticity involving mechanisms of long-term potentiation (LTP) (Huang et al., 2017 [↗](#); Jannati et al., 2023 [↗](#)). On the other hand, tACS causes rhythmic fluctuations in neuronal membrane potentials, which can bias spike timing, leading to an entrainment of the neural activity (Wischnewski et al., 2023 [↗](#)). In particular, the induction of gamma oscillatory activity has been proposed to play an important role in a type of LTP known as spike timing-dependent plasticity, which depends on a precise temporal delay between the firing of a presynaptic and a postsynaptic neuron (Griffiths and Jensen, 2023 [↗](#)). Both LTP and gamma oscillations have a strong link with memory processes such as encoding (Bliss and Collingridge, 1993 [↗](#); Griffiths and Jensen, 2023 [↗](#); Rossi et al., 2001 [↗](#)), pointing to rTMS and tACS as good candidates for memory enhancement.

However, despite the increasing expectations, their overall effects are relatively modest, even with prolonged stimulation sessions (Grover et al., 2022 [↗](#); Polanía et al., 2018 [↗](#); Wang et al., 2014 [↗](#)). In fact, the behavioral and clinical effects are extremely variable (Corp et al., 2020 [↗](#)) especially when used to modulate memory functions (Pabst et al., 2022 [↗](#)).

Recent studies explored the possibility of combining these two techniques showing that the simultaneous application of tACS in the range of the gamma frequency band (ytACS) with intermittent theta burst (iTBS), an rTMS protocol known to induce LTP (Huang et al., 2005 [↗](#)), enhances the after-effects of iTBS alone on the motor and prefrontal cortices (Guerra et al., 2018 [↗](#); Maiella et al., 2022 [↗](#)).

Long-term memory formation has been associated with the neural activity of the precuneus (PC). The PC represents one of the main hubs of the default mode network (Cavanna and Trimble, 2006 [↗](#); Cunningham et al., 2017 [↗](#); Jitsuishi and Yamaguchi, 2021 [↗](#)) and is involved in the formation of associative, episodic, and autobiographic memories (Bonni et al., 2015 [↗](#); Brodt et al., 2018 [↗](#); Flanagan et al., 2023 [↗](#)). Moreover, it is connected with the temporal lobe and the hippocampus (Cunningham et al., 2017 [↗](#); Dadario and Sughrue, 2023 [↗](#); Tanglay et al., 2022 [↗](#)). Hence the PC represents an ideal target for NIBS to modulate memory functions with profound clinical implications such as in the case of Alzheimer's disease (2).

Here we hypothesized that ytACS would activate key neural mechanisms underlying memory formation favoring the iTBS-mediated plasticity induction by entraining and synchronizing endogenous gamma oscillations, thereby improving long-term memory. Thus, we reasoned that

the dual application of iTBS and ytACS over the PC-hippocampus network could be a promising tool for memory enhancement by enhancing plasticity mechanisms.

Results

Dual precuneus iTBS+ytACS improves long-term associative memory

In the first experiment, subjects ($n=20$) were asked to perform two tasks measuring long and short-term memory performances immediately after three balanced, randomized, and double-blind stimulation conditions with a cross-over design (iTBS+ytACS, iTBS+sham-tACS, sham-iTBS+sham-tACS). Stimulation parameters were personalized and identified using fMRI and TMS-EEG (see **Fig. 1**, panels A and B). Long-term memory was assessed by the face-name associative task (FNAT), requiring the memorization of 12 faces with corresponding names and occupations, while short-term memory was assessed by the short-term memory binding test (STMB) (see **Fig. 1**, panel C).

The different stimulation protocols were all well-tolerated. No one reported significant side effects connected with the neuromodulation protocol applications. We found that dual iTBS+ytACS improved long-term associative memory as compared to the other conditions. Dual iTBS+ytACS increased the performances in recalling the association between face, name and occupation (FNAT accuracy) both for the immediate ($F_{2,38}=7.18$; $p=0.002$; $\eta^2_p=0.274$) and the delayed ($F_{2,38}=5.86$; $p=0.006$; $\eta^2_p=0.236$) recall performances (**Fig. 2**, panel A). The sensitivity analysis showed a minimal detectable effect size of $\eta^2_p=0.215$ with 20 participants. Post-hoc analysis showed an advantage for dual iTBS+ytACS as compared to iTBS+sham-tACS condition for the immediate ($40.0\pm 21.6\%$ vs. $25.0\pm 17.9\%$, $p=0.002$ Bonferroni corrected, corresponding to a relative improvement of 63.79%) and for the delayed recall ($34.2\pm 20.4\%$ vs. $24.2\pm 19.8\%$, $p<0.05$ Bonferroni corrected, corresponding to a relative improvement of 34.16%), as well as compared to the sham-iTBS+sham-tACS condition for the delayed recall ($34.2\pm 20.4\%$ vs. $26.3\pm 17.6\%$ $p=0.037$ Bonferroni corrected, corresponding to a relative improvement of 34.04%) (**Fig. 2**, panel A).

The in-depth analysis of the FNAT accuracy investigating the specific contribution of face-name and face-occupation recall revealed that dual iTBS+ytACS increased the performances in the association between face and name (FNAT NAME) delayed recall ($F_{2,38}=3.46$; $p=0.042$; $\eta^2_p=0.154$; iTBS+ytACS vs. sham-iTBS+sham-tACS: $42.9\pm 21.5\%$ vs. $33.8\pm 19\%$; $p=0.048$ Bonferroni corrected) (**Fig. S4**, supplementary information). iTBS+sham-tACS did not result in any improvement as compared to sham-iTBS+sham-tACS. No significant effects were found over the recall of the association between face and occupation (FNAT OCCUPATION) ($p<0.05$). Moreover, we found that the effects of dual iTBS+ytACS stimulation were selective for long-term but not short-term memory, since no effects were found on STMB accuracy and RTs (all $p>0.05$) (**Fig. 2**, panel B). Table S1 (supplementary information) shows Experiment 1 statistical details. Biophysical modeling calculations showed that the iTBS induced a mean e-field of 29.55 ± 5.17 V/m, while ytACS led to a much smaller mean e-field of 0.1 ± 0.55 V/m.

Enhanced Memory is still evident after one week

To confirm the above-described effects of dual iTBS+ytACS on an independent sample ($n=10$) and to investigate the time course of memory retention we performed a second experiment. In this experiment, subjects were asked to execute the delayed FNAT cued recall 24 hours and a 1-week after the neuromodulatory protocols. As for experiment 1, we confirmed that dual iTBS+ytACS improved immediate recall at TOTAL FNAT as compared to iTBS+sham-tACS ($F_{1,9}=7.31$; $p=0.024$; $\eta^2_p=0.448$; $26.7\pm 10.2\%$ vs. $17.5\pm 8.3\%$; $p=0.024$ corresponding to a relative improvement of 82.50%) (see Table S2 in supplementary information for statistical details). The sensitivity analysis showed a minimal detectable effect size of $\eta^2_p=0.388$ with 10 participants. More importantly, we found that

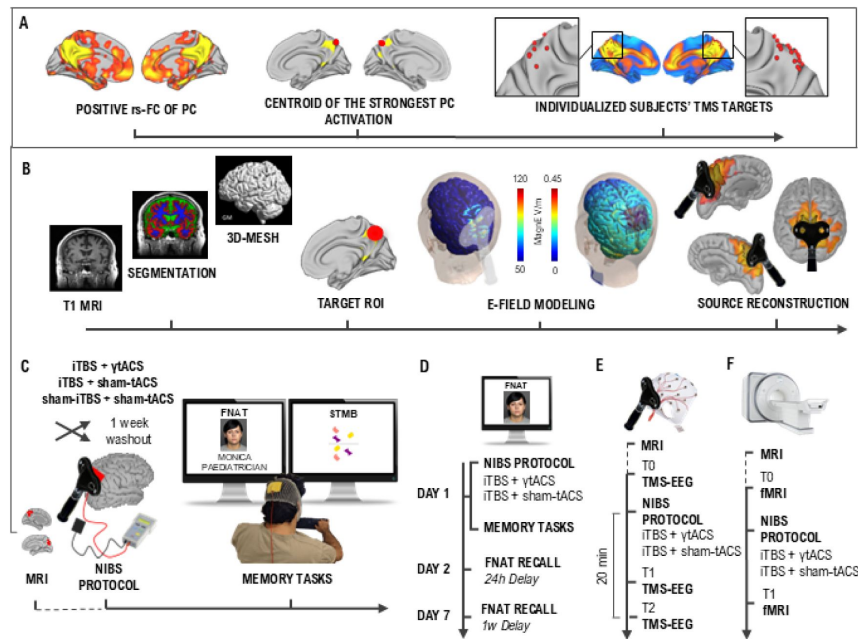


Figure 1.

Individual target extraction and experimental design.

(A) Methodological workflow used to extract the individualized target for the neuromodulation protocol used in experiments 1, 3 and 4. Participants underwent fMRI scanning to individualize the stimulation sites and permit neuronavigation. Target individualization was derived by computing a PC functional connectivity profile for each participant, thus obtaining a map of positively correlated voxels, respectively representing the DMN (Panel A, left). The individual stimulation targets were defined as the centroid of the strongest PC activation being on the top of a cortical gyrus and representing the shortest perpendicular path connecting the stimulating TMS coil on the scalp and the cortex (Panel A, center). PC coordinates for each participant are represented in red on an MNI brain template, showing the overlap with the DMN (Panel A, right). (B) Biophysical modeling was computed for each subject acquiring T1w and T2w MRI and using the Simnibs toolbox for the T1w segmentation and the 3D-mesh transformation (Panel B, left). The mean Norm E-field was extracted from a target ROI-sphere (10 mm radius) centered on the individual coordinates of the PC (Panel B, center). The simulated induced electric field is shown for a representative subject produced by iTBS (e-field modeling, left) and tACS (e-field modeling, right). EEG source activity reconstruction induced by the TMS pulse over the precuneus in a representative subject (Panel B, right). Experimental design of experiments 1 (C), 2 (D), 3 (E), 4 (F). The effect of simultaneous iTBS+ytACS on memory performances was investigated in experiments 1 (C) and 2 (D) through two memory tasks: the face-name associative task (FNAT), which required the memorization of 12 faces with corresponding faces and occupations, and the visual short-term memory binding test (STMB), which consisted in a change detection task. In the main experiment 1 (C), subjects were involved in a cross-over design with different experimental sessions of neuromodulation separated by a washout week. Every session corresponded to a different balanced and randomized stimulation condition (i.e., iTBS+ytACS, iTBS+sham-tACS, sham-iTBS+sham-ytACS) immediately followed by the FNAT learning phase and immediate recall, the STMB and the FNAT delayed recall (15-minute delayed) and recognition.

In experiment 2 (D) subjects were involved in a cross-over design with two balanced and randomized stimulation conditions (i.e., iTBS+ytACS, iTBS+sham-tACS) separated by a washout week. During the first session (day 1), participants received the neuromodulation protocol and then performed the learning phase and immediate recall FNAT, the STMB and the FNAT delayed recall. In the second session (day 2), participants performed FNAT recall with a 24-hour delay from the neuromodulation protocol, while in the third session (day 7), participants performed FNAT recall and recognition with a 1-week delay. In experiment 3 (E) participants were involved in two randomized and balanced experimental sessions of neuromodulation (i.e., iTBS+ytACS, iTBS+sham-tACS) separated by a washout week. TMS-EEG recordings were performed before (T0), immediately after (T1), and 20 minutes after the neuromodulation (T2). In experiment 4 (F), after the first MRI scanning used for neuronavigation, participants were involved in two randomized and balanced experimental sessions of neuromodulation (i.e., iTBS+ytACS, iTBS+sham-tACS) separated by a washout week. The fMRI scanning was performed before (T0) and immediately after (T1) the neuromodulation protocol.

Photographs reported represent the author Michele Maiella performing the task and an example of the face item used in the task taken from the FACES database (Ebner et al., 2010).

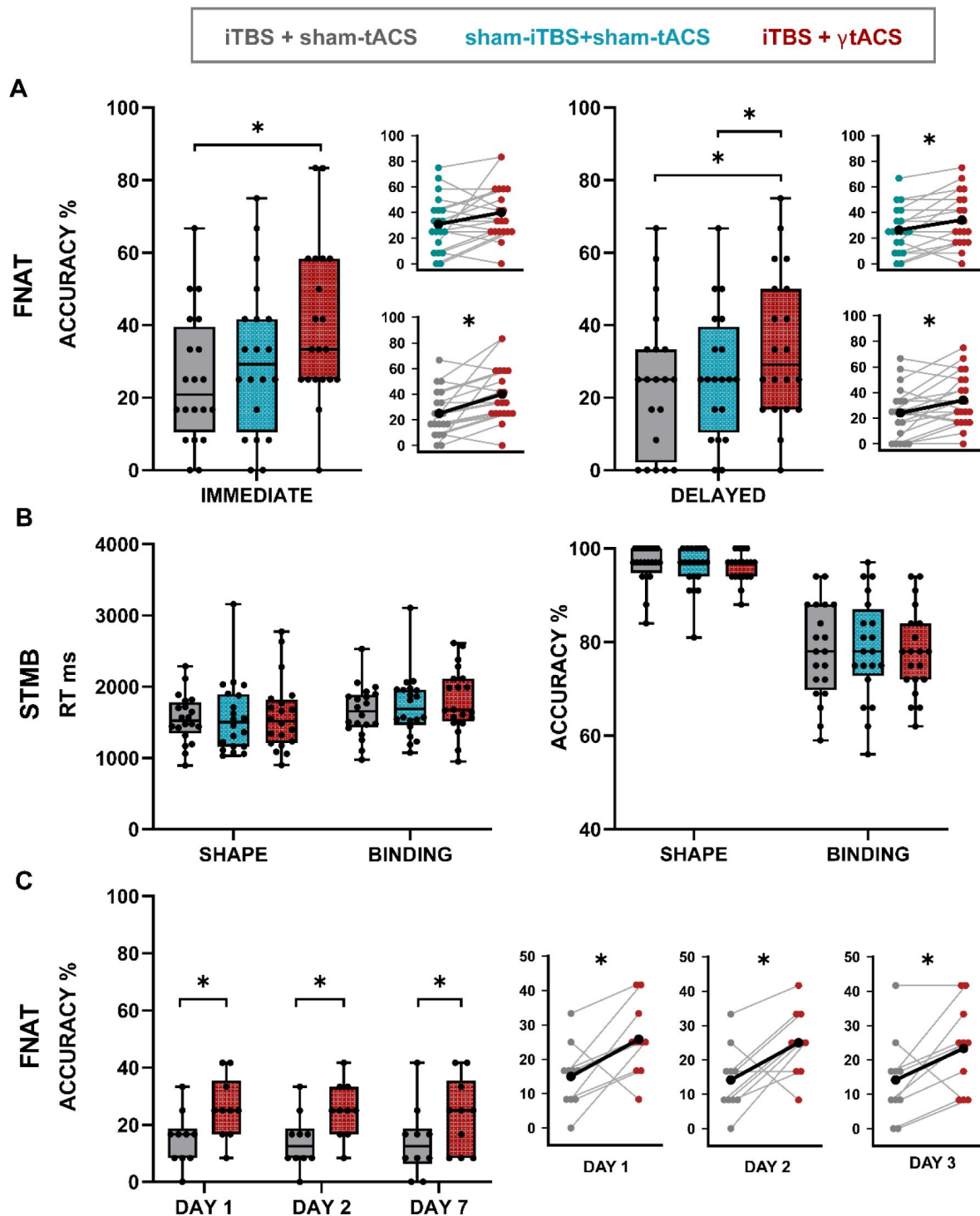


Figure 2.

Memory performance outcome.

In the Box and Whiskers plots, boxes delimit the lower (Q1) and the upper (Q3) quartile, the whiskers extend to the smallest and largest values, dots represent individual values. In the line graphs, colored dots show individual participant data, lines show the connection between the performances of each participant, black dots and lines correspond to the mean performance values. Grey corresponds to iTBS+sham-tACS, blue to sham-iTBS+sham-tACS and red to iTBS+ γ tACS. * = $p < 0.05$

(A) Face-name associative task (FNAT) accuracy in immediate (right) and delayed (left) trials resulting from experiment 1. N=20. (B) Short-term memory binding task RTs (left) and accuracy (right) resulting from experiment 1. N=20. (C) FNAT's long-lasting effect resulted from experiment 2. The results are shown over time (day 1, day 2, day 7). N=10.

dual iTBS+ytACS exerted a long-lasting effect on long-term associative memory ($F_{1,9}=8.433$; $p=0.017$; $\eta^2_p=0.484$) (**Fig. 2**, panel C). The sensitivity analysis showed a minimal detectable effect size of $\eta^2_p=0.125$ with 10 participants. We observed that the incremental effect on TOTAL FNAT performance was still evident at 24 hours and one week after the neuromodulation (day 1: $25.8\pm10.7\%$ vs. $15\pm9.5\%$, $t_9=-2.75$, $p=0.011$ corresponding to a relative improvement of 78.33%; day 2: $25\pm9.6\%$ vs. $14.2\pm9.7\%$, $t_9=-2.89$, $p=0.009$ corresponding to a relative improvement of 88.33%; day 7: $23.3\pm12.9\%$ vs. $14.2\pm12.5\%$, $t_9=-2.40$, $p=0.020$ corresponding to a relative improvement of 80%) (see Table S3 in supplementary information). No significant effects were found for STMB accuracy and RTs (all $p>0.05$) as for experiment 1 (see Table S2 in supplementary information for statistical details).

Precuneus iTBS+ytACS increases gamma oscillatory activity

We then aimed to investigate the effects of the neuromodulation protocol on cortical oscillations (TRSP) and excitability (TEPs), through TMS-EEG recordings ($n=14$ subjects, 6 of those also participated in experiment 1). We found that cortical oscillations in the gamma band increased after dual iTBS+ytACS as compared to baseline TMS-EEG recordings (stimulation $F_{1,13}=5.073$; $p=0.042$; $\eta^2_p=0.281$) (**Fig. 3**, panel A). Specifically, we observed an increase of the gamma-TRSP after dual iTBS+ytACS as compared to iTBS+sham-tACS in both time points ($\Delta T1$: 0.045 ± 0.09 vs. -0.013 ± 0.06 ; $t_{13}=1.96$; $p=0.036$; $\Delta T2$: 0.026 ± 0.04 vs. -0.013 ± 0.05 ; $t_{13}=1.92$; $p=0.039$) (**Fig. 3**, panel B). This effect was specific to PC; indeed, no significant effect was found when testing TMS-EEG over the l-PPC ($p>0.05$). This result was reflected by increased cortical excitability evident after dual iTBS+ytACS but not after iTBS+sham-tACS condition (**Fig. 3**, panel C). Specifically, we observed a significant TEP amplitude increase over a cluster of parietal and occipital electrodes comparing T0 vs T1 (mean $t\text{-value}_{13}=-3.29$; all $p<0.05$) and over POz ($t\text{-value}_{13}=-4.74$; $p<0.05$) when comparing T0 vs. T2. No significant differences were found comparing T0 vs. T1 and T2 in the iTBS+sham-tACS condition ($p>0.05$).

Precuneus iTBS+ytACS increases connectivity with the hippocampus

Finally, we investigated the effects of the dual iTBS+ytACS protocols on MRI-based resting state functional connectivity ($n=16$ subjects, 7 of whom participated in experiment 1). ROI-to-ROI analysis revealed a significant difference specifically for the connectivity between the PC and the hippocampi ($F_{3,13}=5.78$, $p\text{-FDR}=0.029$). Post-hoc analysis showed increased connectivity after the stimulation for the dual iTBS+ytACS ($t_{15}=3.67$, $p\text{-FDR}=0.0023$; **Fig. 4**, panel A), but not for the iTBS+sham-tACS. No significant differences were detected among the conditions before stimulation (all $p>0.05$). The significant nodes that emerged from the ROI-to-ROI analysis (i.e. PC and hippocampi) were considered for the seed-to-voxel analyses, revealing an increased rs-FC between the PC and the orbitofrontal cortex) after dual iTBS+ytACS as compared to baseline (OFC; MNI coordinates: $x=20$, $y=16$, $z=-4$; $|T15| > 3.20$, $k=1051$; $p=0.0003$) (**Fig. 4**, panel B center). Moreover, stronger connectivity with the PC (MNI coordinates: $x=30$, $y=-70$, $z=46$; $|T15| > 3.20$, $k=1499$; $p=0.000167$) and with the posterior cingulate cortex emerged when considering the right hippocampus as seed (MNI coordinates: $x=-14$, $y=-50$, $z=12$; $|T15| > 3.20$, $k=1295$; $p=0.0001649$) (**Fig. 4**, panel B right). Finally, similar results occurred considering as seed the left hippocampus, with an improved rs-FC with the PC (MNI coordinates: $x=30$, $y=-68$, $z=44$; $|T15| > 3.20$, $k=1525$; $p=0.000163$) and with the posterior cingulate cortex (MNI coordinates: $x=20$, $y=-50$, $z=12$; $|T15| > 3.20$, $k=509$; $p=0.022$) (**Fig. 4**, panel B left). We also reconstructed the white matter fibers connecting the PC with the temporal lobe in each subject, forming the bilateral Middle Longitudinal Fasciculus (MdLF), using DTI tractography. We found that the structure of the MdLF measured through fractional anisotropy (FA) was related to the increased connectivity between PC and the hippocampi after dual iTBS+ytACS. A significant positive correlation emerged between the FA of the MdLF and the changes in connectivity between PC and hippocampus in the iTBS+ytACS ($r=0.73$, $p=0.001$), but not in the iTBS+sham-tACS ($r=0.14$, $p=0.634$) (**Fig. 4**, panel C).

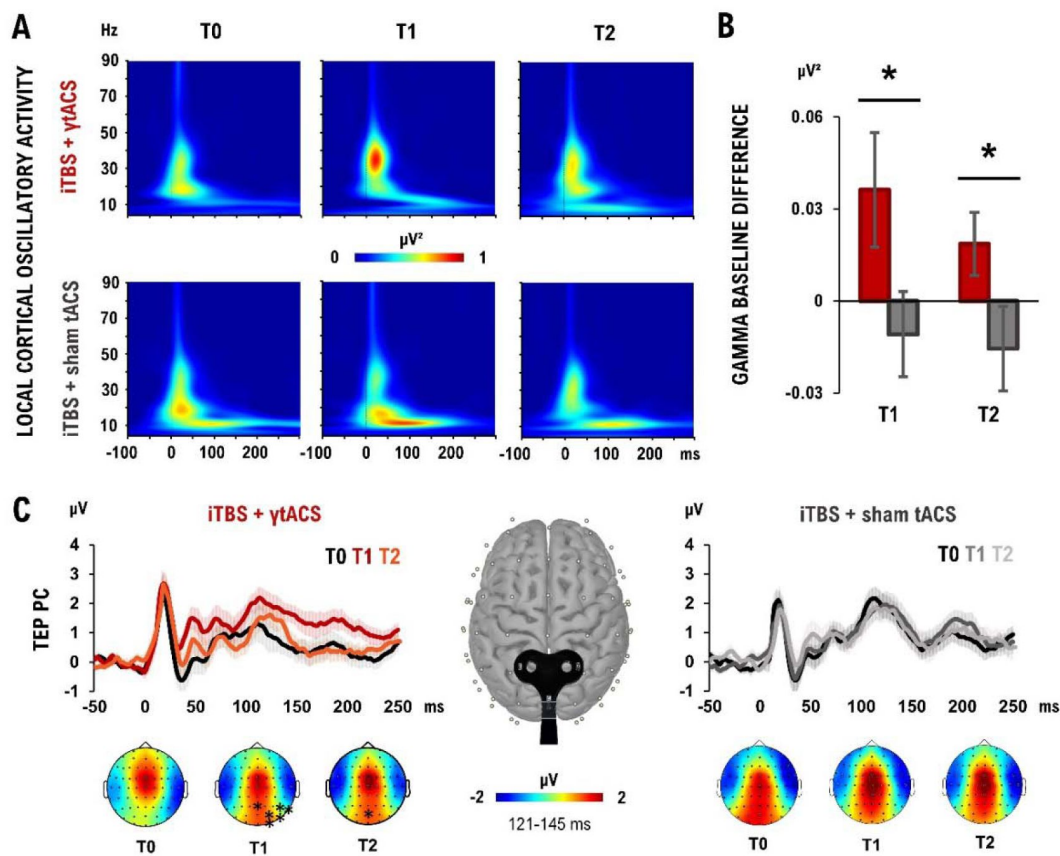


Figure 3.

Experiment 3 neurophysiological outcome.

(A) Precuneus (PC) oscillatory activity elicited by iTBS+ytACS (up) and iTBS+sham-tACS (down) when testing PC over the three time points (T0, T1, T2 from left to right). (B) Gamma oscillation changes from baseline after iTBS+ytACS (red) and iTBS+sham-tACS (grey). (C) TMS-evoked potential (TEP) produced over the PC when performing TMS-EEG over PC in the two stimulation conditions: iTBS+ytACS (up-left) and iTBS+sham-tACS (up-right) over the three time points (T0, T1, T2). (Down) Topographies and statistical differences in TEPs amplitude after the different stimulation conditions (iTBS+ytACS, left; iTBS+sham-tACS, right) over the three time points (T0, T1, T2, from left to right). N=14; * = $p < 0.05$; bars depict standard error.

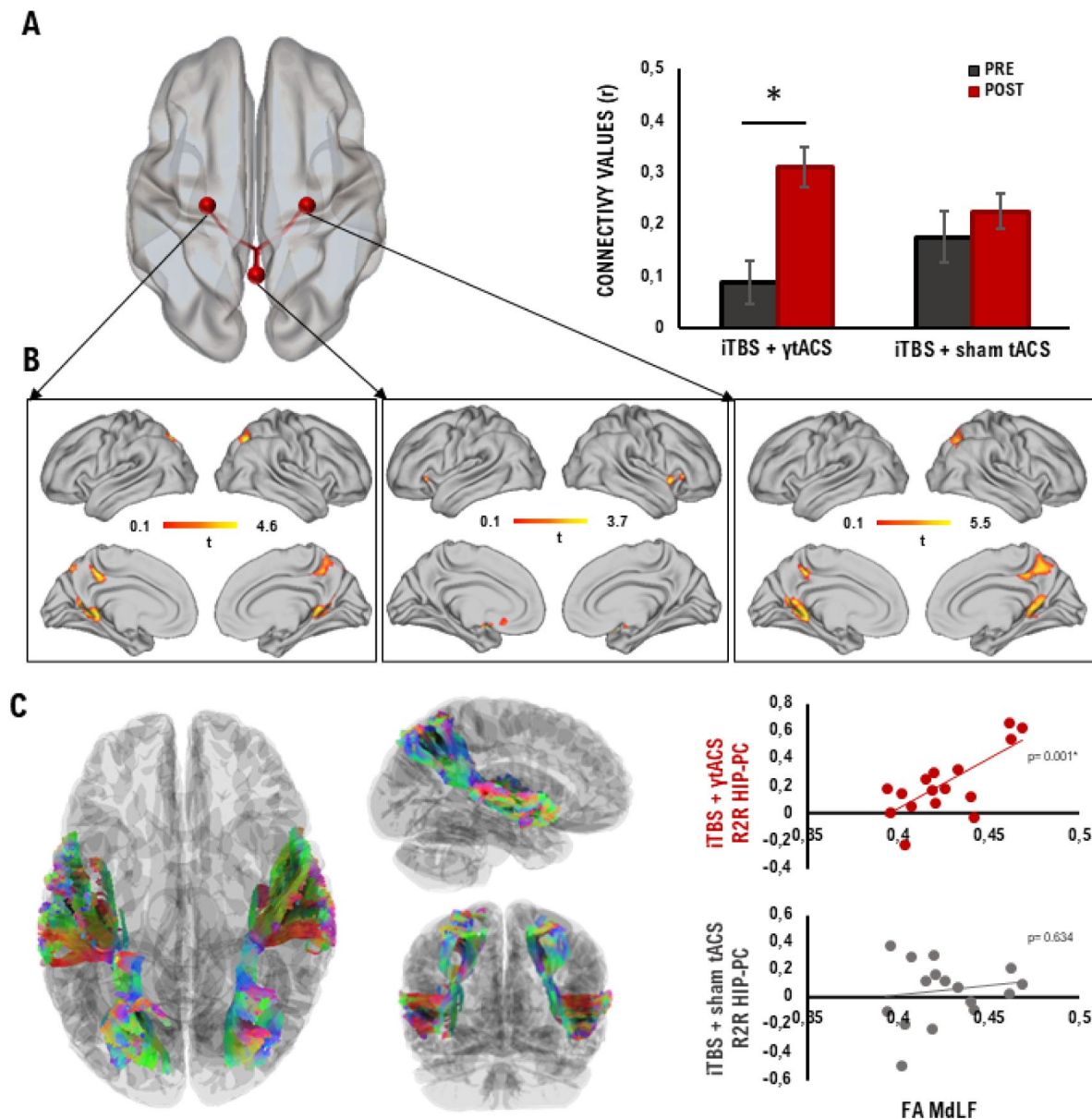


Figure 4.

rsFC changes after iTBS+ytACS and correlation with DTI.

(A) The standard MNI brain (left) and the bar plot (right) display the positive correlation between the PC and the bilateral HIP after the iTBS+ytACS resulted from the ROI-to-ROI analysis. (B) Seed-to-voxel analysis results from each significant ROI (i.e. left HIP, left PC, center; right HIP, right) are overlaid to a standard MNI brain. (C) MdlF extracted (left); positive correlation between MdlF integrity and functional connectivity changes between the PC and bilateral HIP after iTBS+ytACS (up right); the absence of correlation in the iTBS+sham-tACS condition and functional connectivity (down right). N=16; * = $p < 0.05$; bars depict standard error.

Discussion

Here we demonstrated that the personalized combination of iTBS and γ TACS applied over the PC can enhance long-term associative memory formation. Dual stimulation increased the ability to recall the association between faces, names, and occupations as compared to the iTBS alone or sham conditions. Moreover, the increased memory trace was long-lasting, being evident one week after the stimulation. On the other hand, iTBS alone did not result in any memory improvement, in agreement with previous studies showing contradictory effects of this protocol on neuroenhancement (Hamada et al., 2013 [↗](#); Ziemann and Siebner, 2015 [↗](#)). Hence, we show that the combination of neuroplasticity induction promoted by iTBS and gamma oscillations entrainment produced by γ TACS, is the key to memory boosting. We believe that the novel results reported here may have several implications for treating memory impairments, gamma dysregulation, and network dysfunction in a large variety of conditions.

Our study identifies a central role for the PC in mediating long-term memory formation. Given the existing evidence on TMS propagation and the computation of the Biophysical model with the Efield, we can reasonably assume that the individually identified PC was a mediator of the observed effects (Ridding and Rothwell, 2007 [↗](#)). Moreover, we observed specific cortical changes in the posteromedial parietal areas, as evidenced by the whole-brain analysis conducted on TMS-EEG data and the absence of effect on the lateral posterior parietal cortex used as a control condition. The PC has been associated with memory engram formation and retrieval (Brodt et al., 2018 [↗](#); Flanagan et al., 2023 [↗](#); Hebscher et al., 2019 [↗](#)). Moreover, it is a prime candidate in processing relations among entities, including semantic concepts, making its activity fundamental for associative memory formation (Fernandino et al., 2022 [↗](#); Rentz et al., 2011 [↗](#); Summerfield et al., 2020 [↗](#)). These notions found support in our current findings, showing that neuromodulation of the PC resulted in robust improvement of long-term associative memory.

Visual short-term associative memory, measured by STBM performance, was not modulated by any experimental condition. Even if we cannot exclude the possibility that the stimulation could have influenced short-term verbal associative memory, we expected this result since short-term associative memory is known to rely on a distinct frontoparietal network while FNAT, used to investigate long-term associative memory, has already been associated with the neural activity of the PC and the hippocampus (Parra et al., 2014 [↗](#); Rentz et al., 2011 [↗](#)). Moreover, we observed a main effect on the ability to concomitantly recall face, name and occupation, while for single association we observed minimal (face-name) or no effect (face-occupation). Forming face-name associations is widely acknowledged to be particularly difficult due to the lack of contextual properties to formulate an associative link (Werheid and Clare, 2007 [↗](#)). In contrast, forming an association between a face and other biographical information, such as occupations or hobbies, is easier (McWeeny et al., 1987 [↗](#)). We can hypothesize that retaining multiple associations (face-name-occupation) together is even more effortful and cognitively demanding. For these reasons, the stimulation could have promoted effects in a gradient way related to the difficulty of the task. Apart from improving long-term memory, dual iTBS+ γ TACS promoted a long-term increase in gamma oscillatory activity, as measured by TMS-EEG. Cortical gamma-band oscillations are thought to be primarily generated by parvalbumin-positive GABA-ergic interneurons interacting with excitatory pyramidal cells (Buzsáki and Wang, 2012 [↗](#); Cobb et al., 1995 [↗](#)). The repeated activation of these circuits through gamma oscillations fosters LTP formation which is the neurobiological basis of long-term memory (Bikbaev and Manahan-Vaughan, 2008 [↗](#); Debanne and

Inglebert, 2023; Headley and Weinberger, 2011 [↗](#)). This finding is consistent with the idea that increased gamma oscillatory activity may facilitate cortical plasticity, neural communication, and mechanisms of memory encoding and retrieval, thereby ensuring the successful formation of an

episodic memory (Griffiths and Jensen, 2023 [↗](#)). We propose that ytACS, by entraining and synchronizing ongoing oscillatory gamma activity, may have facilitated the formation of LTP induced by iTBS.

Furthermore, we found that dual iTBS+ytACS enhanced the connections between the PC and the hippocampus as measured by fMRI connectivity analysis. Effective neural communication driven by gamma oscillations ensures that, during memory formation, incoming information processed in the neocortex activates the relevant cell assemblies in the hippocampus to ensure associative binding (Griffiths and Jensen, 2023 [↗](#)). Gamma-band oscillatory activity drives network connectivity allowing for information processing and binding through synchronization between distant cortical areas (Fries, 2005 [↗](#); Jensen et al., 2007 [↗](#); Miltner et al., 1999 [↗](#); Osipova et al., 2006 [↗](#)). Hence we argue that PC gamma oscillations would bind relevant stimuli for perceptual representation, while synchronization between cortical and hippocampal neurons would allow the representation to be encoded into the hippocampus (Nyhus and Curran, 2010 [↗](#)).

Finally, we found that higher functional connectivity was supported by higher white matter integrity (FA) of MdLF, the main white matter tract connecting the PC with the temporal lobe. Altogether these findings clarify the anatomo-functional network engagement due to PC stimulation, providing further support for the successful engagement of long-range pathways connecting the PC with the temporal lobe underlying memory formation.

A strength of the current work is the personalization of stimulation parameters that was achieved by targeting the PC spot more connected with the temporal lobe and by selecting the intensity of stimulation for iTBS using TMS-EEG. Hence we argue that personalization of NIBS protocols is a key factor in reducing the inter-individual variability that limit currently the use of common protocols such as TBS and in increasing the overall beneficial effects on cognitive functions.

Why should the dual application of ytACS+iTBS lead to such a strong boosting effect as compared to iTBS alone? It is well established that tACS alone does not induce relevant after-effects, especially when applied for a short period of 3 minutes such as in the current case (Guerra et al., 2019 [↗](#), 2018 [↗](#)). However, during stimulation, tACS alters membrane potential towards depolarization or hyperpolarization in an oscillatory fashion, manipulating the excitability of neurons that become aligned with the introduced electric field, mostly pyramidal cells in layer V (Fröhlich and McCormick, 2010 [↗](#); Guerra et al., 2018 [↗](#); Reato et al., 2013 [↗](#); Schutter and Hortensius, 2011 [↗](#)). Notably, these neurons are characterized by intrinsic resonance and neuroplastic activity. In particular, synchronous neural activation in the γ range is considered crucial for the induction of neuroplasticity and memory formation (Buzsáki and Wang, 2012 [↗](#)). Gamma oscillations can synchronize the firing of multiple presynaptic neurons so that they exert a stronger depolarizing effect on the target postsynaptic neuron than if they were to fire in isolation (Sjöström et al., 2001 [↗](#)). Synchronizing multiple inputs to a postsynaptic neuron enhances the likelihood of LTP. Hence, gamma oscillations are perhaps ideal because they provide a comparatively short window of excitability that ensures all neurons fire in near-perfect unison (Jensen et al., 2007 [↗](#)). This provides a long-sought link between gamma oscillations, LTP, and the formation of new memories. For these reasons, ytACS may have provided the ideal neural substrate for the iTBS to exert its plasticity induction mechanisms. This might have occurred by driving neuronal assemblies in a state devoted to inducing long-term changes in cortical networks involved in memory formation. In humans, iTBS has been inspired by protocols used in animal models to elicit LTP in the hippocampus (Huang et al., 2005 [↗](#)).

On top of iTBS, ytACS may have synchronized PC neuronal elements boosting gamma oscillatory activity during the induction of LTP driven by iTBS. Notably, this interaction is rather specific, since previous studies showed that the application of ytACS alone does not induce any after-effects, while only gamma, but not theta or beta tACS, combined with iTBS induces long-lasting changes in cortical excitability (Guerra et al., 2018 [↗](#); Maiella et al., 2022 [↗](#)). Moreover, our

biophysical modeling calculations showed that ytACS induced a marginal augmentation of the e-field (~ 0.15 mV) which is below the amount of induced current needed to induce an effect in superficial cortical neurons (Opitz et al., 2016). Hence, tACS alone cannot account for the observed increase in memory as well as for enhancing cortical activity.

This study has some limitations. Although we studied TMS and tACS propagation through the E-field modeling and observed an increase in the precuneus gamma oscillatory activity, excitability and connectivity with the hippocampi, we cannot exclude that our results might reflect the consequences of stimulating more superficial parietal regions other than the precuneus nor report direct evidence of microscopic changes in the brain after the stimulation. Invasive neurophysiological recordings in humans for this type of study are not feasible due to ethical constraints. Studies on cadavers or rodents would not fully resolve our question due to significant differences between them (i.e. rodents do not have an anatomical correspondence while cadavers have an alterations in electrical conductivity occurring in postmortem tissue). However, further exploration of this aspect in future studies would help in the understanding of ytACS+iTBS effects.

We did not assess the effects of ytACS alone. This decision was based on the findings of Guerra et al. (Guerra et al., 2018), who investigated the same protocol and reported no aftereffects. Given the substantial burden of the experimental design on patients and our primary goal of demonstrating an enhancement of effects compared to the standalone iTBS protocol, we decided to leave out this condition. While examining the effects of ytACS alone could help isolate its specific contribution to this target and memory function, extensive research has shown that achieving a cognitive enhancement aftereffect with tACS alone typically requires around 20–25 minutes of stimulation (Grover et al., 2023). We did not study memory functions, gamma oscillations, and synchronization between the PC and hippocampus in the same session due to technical limitations with the current techniques (i.e., fMRI, TMS-EEG). Consequently, these findings do not allow precise inferences regarding the specific mechanisms by which dual iTBS and ytACS of the precuneus modulate learning and memory. Methodological restraints related to the high-frequency protocols employed in this study did not allow for closed-loop implementation. Moreover, we did not test tACS applied at different frequency bands combined with iTBS since our working hypothesis was based on the combination of gamma activity and plasticity induction. Future studies should further investigate the effects of stimulation on distinct memory processes. In particular, stimulation could be applied before retrieval (Rossi et al., 2001), to better elucidate its specific contribution to the observed enhancements in memory performance. Additionally, it would be worth examining whether repeated stimulation - administered both before encoding and before retrieval - could produce a boosting effect. This is especially relevant in light of findings showing that matching the brain state between retrieval and encoding can significantly enhance memory performance (Javadi et al., 2017). Our findings have profound social and clinical implications.

These results may assume relevance in the context of memory impairment, which is an increasing and challenging aspect of the elderly population. The current findings may be relevant in the case of Alzheimer's disease, where the progressive memory deficit is accompanied by early involvement of DMN with an accumulation of beta-amyloid plaques, neurofibrillary tangles, atrophy, and functional connectivity primarily targeting the PC (Billette et al., 2022; Buckner et al., 2005; Chen et al., 2017; Raichle et al., 2001). In this perspective, the DMN has been recently identified as a new potential therapeutic target for neuromodulation in AD (2). Moreover, by linking together items and concepts, long-term associative memory is essential for the construction of declarative memory and, broadly for learning and everyday functioning. Hence, dual magnetic and electrical transcranial stimulation could be promising in several social and clinical contexts such as learning deficits, autism spectrum, and attention deficit hyperactivity disorders (Mathalon and Sohal, 2015). Moreover, future studies should investigate the effect of the co-stimulation when administrated prior to recall to clarify the specific

Materials and Methods

Experimental design

We conducted four experiments aimed at understanding the effects of dual iTBS+ytACS on associative memory performance, cortical oscillatory activity and reactivity, and functional connectivity.

Experiment 1. We investigated the effect of dual iTBS+ytACS on memory performances through two memory tasks: the face-name associative task (FNAT) and the visual short-term memory binding test (STMB). Participants first underwent MRI scanning to individualize the stimulation sites and permit neuronavigation (**Fig. 1** [↗](#), panel A), then they were involved in a double-blind cross-over design with three experimental sessions of neuromodulation separated by a washout week (**Fig. 1** [↗](#), panel C). Every session corresponded to a different balanced and randomized stimulation condition (i.e., iTBS+ytACS, iTBS+sham-tACS, sham-iTBS+sham-ytACS) immediately followed by the two memory tasks. ytACS alone was not administrated since a single 3-minute ytACS does not exert any relevant after-effect (Guerra et al., 2018 [↗](#)).

Experiment 2. We deepen the study of memory effects by confirming experiment 1 results and investigating the course of the memory trace over time after the neuromodulation protocol. We simplified the experimental design by excluding one of the control conditions, i.e. ‘sham-iTBS+sham-ytACS’, as we did not observe any effect related to this condition (see ‘results’ paragraph). In particular, participants were involved in a double-blind cross-over design with two balanced and randomized stimulation conditions (i.e., iTBS+ytACS, iTBS+sham-tACS) separated by a washout week. Participants were required to attend three experimental sessions for each condition. During the first session (day 1), participants received the neuromodulation protocol and then performed FNAT and STMB. In the second session (day 2), participants performed FNAT recall with a 24-hour delay from the neuromodulation protocol, while in the third session (day 7), participants performed FNAT recall and recognition with a 1-week delay (**Fig. 1** [↗](#), panel D).

Experiment 3. We investigated the neurophysiological effects of the neuromodulation protocol through the combined use of TMS and electroencephalography (TMS-EEG). As for experiment 1, participants underwent an MRI scanning to individualize the stimulation sites and permit neuronavigation. Then, they were involved in two balanced, double-blind, randomized experimental sessions of neuromodulation (i.e., iTBS+ytACS, iTBS+sham-tACS) separated by a washout week. TMS-EEG recordings were performed before (T0), immediately after (T1), and 20 minutes after the neuromodulation (T2) (**Fig. 1** [↗](#), panel E).

Experiment 4. We investigated the effects of the neuromodulation protocol on functional connectivity through fMRI scanning. After the first MRI scanning used for neuronavigation, participants were involved in two balanced, double-blind, randomized experimental sessions of neuromodulation (i.e., iTBS+ytACS, iTBS+sham-tACS) separated by a washout week. The fMRI scanning was performed before (T0) and immediately after (T1) the neuromodulation protocol (**Fig. 1** [↗](#), panel F).

Participants

We recruited young healthy participants who provided written informed consent approved by the Santa Lucia Foundation IRCCS ethical committee (CE/PROG.923) in accordance with the Declaration of Helsinki. See supplementary information for inclusion and exclusion criteria.

In total 41 subjects were enrolled. 24 participants were involved in experiment 1 in which we tested the effects of the dual iTBS and ytACS on memory functions. Another group of 10 was enrolled in experiment 2 to replicate the results of experiment 1 in an independent sample and to investigate the long-term effect of the dual stimulation. 16 participants were recruited in experiment 3, of which 10 had already participated in experiment 1, to examine the effects on cortical oscillatory activity and cortical reactivity. Finally, 18 participants were involved in experiment 4, of which 12 had already participated in experiment 1, to investigate functional connectivity through fMRI.

Eight participants (i.e., n=4 from experiment 1, n=2 from experiment 3, n=2 from experiment 4) voluntarily withdrew before completing all the experimental sessions. We analyzed at all time points in n=20 subjects for experiment 1, n=10 for experiment 2, n=14 for experiment 3, and n=16 for experiment 4 (see **Table 1** [↗](#) for demographic characteristics). See supplementary information for sample size estimation.

iTBS+ytACS neuromodulation protocol

The co-stimulation consisted of a combination of iTBS and tACS delivered in the gamma band (ytACS) based on previous studies ([Guerra et al., 2018](#) [↗](#); [Maiella et al., 2022](#) [↗](#)). The neuromodulation protocol was delivered on the individual PC, based on the individual resting state structural and functional MRI (see ‘MRI data acquisition and preprocessing’ paragraph), and targeted with a stereotaxic neuronavigation system (SofTaxis, E.M.S., Bologna s.r.l.). The active tACS electrode (anode) was placed on the scalp with the iTBS coil above it, and the other tACS electrode (cathode) was placed over the shoulder’s right muscle (**Fig. 1** [↗](#), panel C).

tACS was delivered through a Brainstim multifunctional system for low-intensity transcranial electrical stimulation (E.M.S., Bologna s.r.l.) and saline-soaked sponge electrodes (7×5 cm²). ytACS sinusoid frequency wave was set at 70 Hz with an intensity of 1mA for a total duration of 190s. iTBS was delivered through a MagStim Rapid² magnetic stimulator (Magstim Company, Whitland, Wales, UK) delivering a biphasic waveform pulse (pulse width ~ 0.1 ms) connected to a figure-of-eight coil (70 mm). iTBS consisted of ten bursts of three pulses at 50 Hz lasting 2 s, repeated every 10 s with an 8 s pause between consecutive trains, for a total of 600 pulses total lasting 190 s ([Huang et al., 2005](#) [↗](#)). Biophysical modeling and E-field calculation was conducted to control for the e-field distribution (see supplementary information). Stimulation parameters were personalized and identified using fMRI, TMS-EEG and electromyography (see supplementary information for details).

No ramp-up and no ramp-down were programmed for the stimulation. Sham stimulation conditions were implemented to control the individual contribution of the techniques and the placebo effect. tACS sham conditions were implemented applying only a 2 s ramp up and 2 s ramp down at the beginning and at the end of stimulation, to give the participant the feeling of real stimulation. Sham-iTBS conditions were implemented by adding a wood layer under the coil ([Sandrini et al., 2020](#) [↗](#)). See supplementary information for randomization and masking information.

Memory tasks

Face-name associative task

In experiments 1 and 2, just after the neuromodulation protocol, the participants underwent the FNAT, a cross-modal associative memory test that requires participants to pair pictures of unfamiliar faces with common first names and occupations ([Papp et al., 2014](#) [↗](#)). See supplementary information for details about the task construction.

	Sample (n)	Age (years)*	Sex (female\male)	Education (years)*
Experiment 1	20	28.6 (4.0)	12\8	18.8 (3.1)
Experiment 2	10	24.1 (4.7)	6\4	17.4 (4.7)
Experiment 3	14	27.1 (2.5)	11\3	19.4 (2.1)
Experiment 4	16	26.9 (2.4)	11\5	18.9 (1.5)

*Mean (s.d.)

Table 1.

Demographic characteristics.

The task consisted of a learning phase followed by a recall and recognition trial. During the learning phase, 12 faces associated with a name and an occupation were shown for 8 seconds on a computer screen. Participants were instructed to read aloud the name and the occupation, to ensure their attention was focused on the items, and to memorize the face-name-occupation association. Right after the learning phase, there was an immediate cued recall phase, in which the participants were asked to recall the name and the occupation of each face previously shown while observing the faces again for 8 seconds each. The delayed cued recall consisted of showing the same faces and asking participants to say each name and occupation. Finally, during the recognition trial, participants were asked to recognize the studied face from a distractor face matched for age and sex. Then, for those names and/or occupations not recollected, they had to identify the correct name and/or occupation among three alternatives each. The distractors were a novel name/occupation and a name/occupation associated with a different face. We considered a correct association when a subject was able to recall all the information for each item (i.e. face, name and occupation), resulting in a total of 36 items to learn and associate. To further investigate the effect on FNAT we also computed a partial recall score accounting for those items where subjects correctly matched only names with faces (FNAT NAME) and only occupations with faces (FNAT OCCUPATION). See supplementary information for score details. In experiment 1, the learning phase was followed by an immediate cued recall and a 15-minute delayed cued recall with recognition (Fig. S2, panel A in supplementary information) while in experiment 2 the learning phase was followed by an immediate cued recall and a 15-minute delayed recall on day 1, a 24-hour delayed cued recall on day 2, and a 1-week delayed cued recall with recognition on day 7 (Fig. S2, panel B in supplementary information). The recognition trial was administered just on the third session to avoid a learning bias over day 2 and day 7 cued recall.

The Short-Term Memory Binding test

This test was run between the FNAT immediate cued recall and the 15-minute delayed cued recall. STMB is a recognition task relying on a change detection paradigm elaborated by Parra and colleagues (Parra et al., 2014 [DOI](#)). Participants were instructed to remember visual arrays of three black shapes (shape only condition, Fig. S3, panel A in supplementary information) or colored shapes (shape-color binding condition, Fig. S3, panel B in supplementary information) presented for 2 seconds (study phase). After a 1-second delay where a blank screen is presented (retention interval), a display with the same or different items appears in new random locations on the screen (test phase). Participants were asked to press the “1” button on the keyboard if the items shown in the study and test phase were different (50% of the trials) or to press “2” if they were the same. A total of 32 randomized trials were presented for each condition. Conditions were counterbalanced across participants. Before starting the test, each participant underwent a perception trial where the two arrays of shapes were presented on the same screen, to exclude perceptual deficit and to train participants at the keyboard answers (Fig. S3, panel C in supplementary information). The test was administered through E-prime 2.0 (Psychology Software Tools, Pittsburgh, USA) which recorded accuracy and reaction times (RTs). We derived two accuracy indexes, respectively for the shape-only condition and the shape-color binding condition (total score of each condition/32×100).

TMS-EEG data acquisition

Neurophysiological effects in cortical oscillations, excitability, and connectivity were assessed using single-pulse TMS during EEG recordings. During the TMS-EEG assessment, participants sat on a comfortable armchair in a soundproof room, were instructed to fixate a black cross on the wall, and wore in-ear plugs that continuously played a white noise to avoid possible auditory event-related potential responses (Rocchi et al., 2021 [DOI](#)). TMS-EEG recordings were performed in a resting state to avoid contamination of cortical activity due to underlying memory tasks. The intensity of the white noise was adjusted individually by increasing the volume (always below 90 decibels) until the participant was sure that s/he could no longer hear the TMS-induced click. TMS for EEG recordings was the same stimulator as for the neuromodulation protocol. The stimulated

areas were PC, the area of interest, and the left posterior parietal cortex (l-PPC), considered as a control area. The order of the stimulation was counterbalanced across participants. PC and l-PPC were both identified by the individual resting state structural and functional MRI (**Fig. 1** [↗](#), panel A; see ‘MRI data acquisition and preprocessing’ paragraph). The coil position was constantly monitored using the Softaxic neuronavigation system (E.M.S. Products, Bologna, Italy) and was differently oriented depending on the area of stimulation so that the direction of current flow in the most effective (second) phase was in the posterior-anterior direction. To target the PC, the coil was positioned with an orientation parallel to the midline, while to target the l-PPC the coil was positioned with an orientation of 15 degrees from the midline (Koch et al., 2018 [↗](#)). The single-pulse TMS intensity was set at 100% eSI separately acquired for individual PC and lPPC (see ‘TBS+ytACS neuromodulation protocol’ paragraph) without the tACS electrode under the coil. Each TMS-EEG block consisted of 90 single pulses with a randomized inter-stimulus interval (ISI) between 2 and 4 s. EEG was continuously recorded from 61 scalp sites positioned according to the 10-20 International System, using TMS-compatible Ag/AgCl pellet electrodes mounted on an elastic cap (BrainAmp; BrainProducts GmbH, Munich, Germany). Additional electrodes were used as a ground and reference: the ground electrode was positioned in Fpz, while the reference was positioned on the tip of the nose. EEG signals were digitized at a sampling rate of 5 kHz. Skin/electrode impedance was maintained below 5 k Ω . See supplementary information for TMS-EEG preprocessing and analysis.

MRI data acquisition

MRI data were acquired (1) before experiments 1, 3 and 4, to identify and individualize the stimulation target and (2) before and after the neuromodulation protocol, to assess functional connectivity effects (**Fig. 1** [↗](#), Panel C).

Imaging was acquired on a Siemens PRISMA scanner with a 64-channel head-coil (Siemens, USA). During the first acquisition (1), both the structural and functional MRI (fMRI) were run, whereas in the following acquisition (2) only the fMRI was acquired.

The structural imaging was acquired using high-resolution T1-weighted (T1w) anatomical images obtained through a 3D-MPRAGE sequence (TR=2500ms, TE=2 ms, TI=1070ms, flip angle (FA)=8°, thickness=1mm, imaging matrix=240 × 240), and a Diffusion Tensor Images (DTI) sequence Diffusion Tensor Images (DTI) sequence (TR=3400 ms, TE=80ms; 121 directions, b-value=1000 s/mm², thickness=1.79 mm, gap=1.79 mm, flip angle=90), only acquired before the experiment 3 and 4. The fMRI images were acquired using standard echo-planar blood oxygenation level-dependent (BOLD) imaging (TR=800ms, TE=3 ms, flip angle (FA)=52°, thickness=2.4mm, gap=2.4mm). Subjects were instructed to not focus their thoughts on any particular topic, to not cross their arms or legs, and to keep their eyes open.

For the individualization of the stimulation sites (**Fig. 1** [↗](#), panel A), the functional region of interest (ROI) representing the PC node of the DMN (for experiments 1, 3 and 4) and the l-PPC node of the fronto-parietal network (FPN) (for experiment 3) was derived from the Harvard-Oxford atlas available in CONN (Whitfield-Gabrieli and Nieto-Castanon, 2012 [↗](#)). A seed-to-voxel correlation map was computed for each participant, thus obtaining a map of positively and negatively correlated voxels, respectively representing the DMN and FPN. An investigator expert in MRI checked both resting-state functional connectivity (rs-FC) maps and structural MRI data (i.e., T1-weighted images) to identify individual hotspots based on ad-hoc criteria. In particular, the stimulation sites were defined as the ones closest to the local maxima of the rs-FC cluster identified as DMN-PC and FPN-PPC being on the top of a cortical gyrus and representing the shortest perpendicular path connecting the stimulating TMS coil on the scalp and the cortex. Based on best judgment, the resulting set of coordinates was picked as the individual stimulation site. The individual set of coordinates was then transformed using a non-linear transformation to reconstruct the targets in individual brain spaces. Lastly, to ensure a coherent intrasession and

intersession stimulation (Cocchi and Zalesky, 2018 [↗](#); Fitzgerald et al., 2009 [↗](#)), the individualized targets were marked in the subject's anatomical MRI and loaded into our neuronavigation system. Both personalized points were used for the TMS-EEG assessment, whereas only the personalized PC was used for the neuromodulation protocol in experiments 1, 3 and 4. See supplementary information for fMRI and DTI data preprocessing.

Statistical analysis

Memory performance and local oscillatory changes data were analyzed through SPSS v22 (IBM, Armonk, NY). Before undergoing parametric or non-parametric statistical procedures, the assumption of normality distribution of data residuals was assessed with Shapiro-Wilk test. The assumption of sphericity was tested with Mauchly test, if this test was significant, we used the Huynh-Feldt correction. The level of significance was set at $\alpha=0.05$.

To assess memory performances in experiment 1, we run repeated-measures ANOVAs with stimulation condition as a within-subject factor (i.e., iTBS+ytACS; iTBS+sham-tACS; sham-iTBS+sham-tACS) for each dependent variable. In detail, a separate ANOVA was conducted for every FNAT measure (i.e., name, occupation, total) on each memory process (i.e., immediate recall, delayed recall, and recognition). Separate ANOVAs were also conducted for STBM accuracy and RTs both for the shape and the binding condition. Post-hoc comparisons were performed with paired t-tests corrected with Bonferroni method.

To confirm the results obtained from experiment 1 and to investigate the long-lasting effect on long-term memory we performed experiment 2 where we first ran a repeated-measures ANOVA with stimulation condition as a within-subject factor (i.e., iTBS+ytACS, iTBS+sham-tACS) for the immediate recall as for experiment 1. Then a two-way repeated measure ANOVA was conducted for FNAT delayed recall performance where stimulation condition (i.e., iTBS+ytACS, iTBS+sham-tACS) and time (i.e., day1, day2, day7) were considered as within-subject factors. Following the results obtained in this analysis (see 'results' paragraph), we conducted exploratory paired t-tests aimed at investigating the effect of stimulation conditions on each time point.

Since we do not have an a priori effect size for experiment 1 and 2, we performed a sensitivity power analysis to ensure that these experiments were able to detect the minimum effect size with 80% power and alpha level of 0.05.

To assess cortical oscillation changes in experiment 3, we computed repeated-measures ANOVAs with stimulation condition (iTBS+ytACS, iTBS+sham-tACS) and time T1-T0, T2-T0 ($\Delta T1$, $\Delta T2$) as within factors for each frequency band. Also, in this case, paired t-tests were conducted to investigate the effect of stimulation conditions on each time point. A one-tail hypothesis was considered given the precise hypothesis of gamma increase deriving from our previous work (Maiella et al., 2022 [↗](#)).

To assess cortical excitability effects, we used multiple paired t-tests. The analysis was run in the four time windows of the TEP waveform (i.e. 10-30 ms, 21-60 ms, 61-120 ms, 121-150 ms after TMS; see paragraph 'Cortical excitability' in 'TMS-EEG data acquisition and preprocessing') comparing T1 and T2 to T0 in each stimulation condition. To reduce the occurrence of type I errors, we used the Monte Carlo method, which computes the estimates of the significance probabilities from two surrogate distributions constructed by randomly permuting the two original conditions data 3000 times. Moreover, p-values were corrected with the false discovery rate (FDR) method considering the number of electrodes and were considered significant when at least 10 successive t-tests reached the significance threshold ($p<0.05$). The same statistical analysis was performed to assess changes in source activation. The time series extracted and analyzed from the ROIs (see paragraph 'source activation' in 'TMS-EEG data acquisition and preprocessing') ranged from -50 to 150 ms

from the TMS pulse. As for excitability analysis, we avoided the occurrence of type I errors by implementing Monte Carlo permutation and considering a significant result when at least 10 successive t-tests reached the significance threshold ($p < 0.05$).

To test resting-state functional connectivity (rsFC) modulation we used a general linear model (GLM). The statistical analyses were carried out using the CONN (v.20b) toolbox and Matlab 2018b software (Mathworks, MA, USA) and were performed considering stimulation conditions (i.e., iTBS+ytACS and iTBS+sham tACS) and time points (i.e., pre and post) as factors. In particular, we performed a ROI-to-ROI analysis focusing on 6 specific ROIs, based on our hypothesis: left and right PPC, medial prefrontal cortex (mPFC), PC, left and right hippocampus, chosen based on Harvard-Oxford (cortical and subcortical) atlas (www.fmrib.ox.ac.uk/fsl/) (Desikan et al., 2006). Specifically, rsFC changes were calculated by computing the Pearson correlation coefficient between the average time series extracted from each individual ROI. Multiple comparisons between ROIs were then performed using a two-sided contrast with a cluster level of $p < 0.05$ FDR-corrected. In addition, the significant nodes were considered as seeds for the voxel-wise analysis, specifically performed comparing the pre and post iTBS+ytACS. Temporal correlations were calculated between these seeds and all other voxels in the brain. Results were computed by applying a cluster-level threshold of $p < 0.05$ FDR-corrected.

The sample size for experiments 3 and 4 was estimated based on a previous study using the same protocol (Maiella et al., 2022). Based on the effect size reported in this work ($\eta^2 = 0.291$), our power analysis estimated that a sample size of 14 patients would be necessary to obtain the same effect size with 80% power and an alpha level of 0.05.

To assess linear relationships between the white matter integrity and the functional connectivity outcomes, we perform a bivariate correlation between FA values extracted from the MdLF and the functional connectivity modulation induced by the stimulation assessed through the ROI-to-ROI analysis. Correlations were computed with Pearson's coefficient (two-tailed).

Data will be available upon rational statement request at https://docs.google.com/spreadsheets/d/1IYQHxXFP2EVT1AwCX-rhndQzPk16Ghb_/edit?usp=sharing&oid=105875083760241337938&rtpof=true&sd=true.

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Additional information

Author Contributions

Conceptualization IB, GK; data curation IB, LM; formal Analysis IB, LM, EPC, MA; funding acquisition GK, EPC; investigation IB, MM, LM, MF, FC, ES; methodology IB, LM, MM, EPC, GK; Project administration IB, GK; resources IB, LM, GK; supervision: IB, EPC, SB, GK; visualization: IB, LM, GK; Writing – original draft IB, LM, GK; writing – review & editing IB, LM, MM, SB, EPC, GK.

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Additional files

Supplementary information [↗](#)

movie 1 [↗](#)

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Reviewer #1 (Public review):

Summary:

The authors make a bold claim that a combination of repetitive transcranial magnetic stimulation (intermittent theta burst-iTBS) and transcranial alternating current stimulation (gamma tACS) causes slight improvements in memory in a face/name/profession task.

Strengths:

The idea of stimulating the human brain non-invasively is very attractive because, if it worked, it could lead to a host of interesting applications. The current study aims to evaluate one such exciting application.

Weaknesses:

- (1) The title refers to the "precuneus-hippocampus" network. A clear definition of what is meant by this terminology is lacking. More importantly, mechanistic evidence that the precuneus and the hippocampus are involved in the potential effects of stimulation remains unconvincing.
- (2) The question of the extent to which the stimulation approach and the stimulation parameters used in these experiments causes specific and functionally relevant neural effects remains open. Invasive recordings that could address this question remain out of the scope of this non-invasive study. The authors conducted scalp EEG experiments in an attempt to address this question using non-invasive methods. However, the results shown in Fig. 3 are unclear. The results are inconsistently reported in units of microvolts squared in some panels (3A, 3B) and in units of microvolts in other panels (3C). Also, there is insufficient consideration of potential contamination by signal components reflecting eye movements, other muscle artifacts, or another volume-conducted signal reflecting aggregate activity inside the brain.
- (3) Figure 3 indicates "Precuneus oscillatory activity ...", but evidence that the activity presented reflects precuneus activity is lacking. The maps shown at the bottom of Figure 3C suggest that the EEG signals recorded with scalp EEG reflect activity generated across a wide spatial range, with a peak encompassing at least tens of centimeters. Thus, evidence that effects specifically reflect precuneus activity, as the paper's title and text throughout the manuscript suggest, is lacking.
- (4) The paper as currently presented (e.g., Figure 3) also lacks rigorous evidence of relevant oscillatory activity. Prior to filtering EEG signals in a particular frequency band, clear evidence of oscillations in the frequency band of interest should be shown (e.g., demonstration of a clear peak that emerges naturally in the frequency range of interest when spectral analysis is applied to "raw" signals). The authors claim that gamma oscillations change because of the stimulation, but a clear peak in the gamma range prior to stimulation is not apparent in the data as currently presented. Thus, the extent to which spectral measurements during stimulation reflect physiological gamma oscillations remains unclear.
- (5) Concerns remain regarding the rigor of statistical analyses in the revised manuscript (see also point 8 below). Figure 3B shows an undefined statistical test with $p < 0.05$. The statistical test that was used is not explained. Also, a description of how corrections for multiple comparisons were made is missing. Figures 3A and 3C are not accompanied by statistics, making the results difficult to interpret. For Figure 4C, a claim was made based on a significant p-value for one statistical test and a non-significant p-value in another test. This is a common statistical mistake (see Figure 1 and accompanying discussion in Makin and Orban de Xivry (2019) Science Forum: Ten common statistical mistakes to watch out for when writing or reviewing a manuscript. eLife 8:e48175).
- (6) In the second question posed in the original review, I highlighted that it was unclear how such stimulation would produce memory enhancement. The authors replied that, in the absence of mechanisms, there are many other studies that suffer from the same problem. This raises the question of placebo effects. The paper does not sufficiently address or discuss the possibility that any potential stimulation effects may reflect placebo effects.
- (7) The third major concern in the original review was the lack of evidence for a mechanism that is specific to the precuneus. Evidence for specific involvement of the precuneus remains

lacking in the revised manuscript. The authors state: "the non-invasive stimulation protocol was applied to an individually identified precuneus for each participant". However, the meaning of this statement is unclear. Specifically, it is unclear how the authors know that they are specifically targeting the precuneus. Without directly recording from the precuneus and directly demonstrating effects, which is outside of the scope of the study, specific involvement of the precuneus seems speculative. Also, it does not seem as though a figure was included in the paper to show how the stimulation protocol specifically targets the precuneus. In their response to the original reviews, the authors state that posterior medial parietal areas are the only regions that show significant differences following the stimulation, but they did not cite a specific figure, or statistics reported in the text, that show this. In any event, posterior medial parietal areas encompass a wide area of the brain, so this would still not provide evidence for an effect specifically involving the precuneus.

(8) Regarding chance levels, it is unfortunate that the authors cannot quantify what chance levels are in the immediate and delayed recall conditions. This makes interpretation of the results challenging. In the immediate and delayed conditions, the authors state that the chance level is 33%. It would be useful to mark this in the figures. If I understand correctly, chance is 33% in Fig. 2A. If this is the case and if I am interpreting the figure correctly: Gray bars for the sham condition appear to be below chance (~20-25%). Why is this condition associated with an accuracy level that is lower than chance?

Cyan bars and red bars do not appear to be significantly different from chance (i.e., 33%), with red slightly higher than cyan. What statistic was performed to obtain the level of significance indicated in the figure? The highest average value for the red condition appears to be around 35%. More details are needed to fully explain this figure and to support the claims associated with this figure.

(9) In the revised version of the paper, the authors did not address concerns associated with the block design (please see question 4d in the original review).

In sum, this study presents an admirable aspirational goal, the notion that a non-invasive stimulation protocol could modulate activity in specific brain regions to enhance memory. However, the evidence presented at the behavioral level and at the mechanistic level (e.g. the putative involvement of specific brain regions) remains unconvincing.

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Reviewer #2 (Public review):

Summary:

The manuscript by Borghi and colleagues provides evidence that the combination of intermittent theta burst TMS stimulation and gamma transcranial alternating current stimulation (ytACS) targeting the precuneus increases long-term associative memory in healthy subjects compared to iTBS alone and sham conditions. Using a rich dataset of TMS-EEG and resting-state functional connectivity (rs-FC) maps and structural MRI data, the authors also provide evidence that dual stimulation increased gamma oscillations and functional connectivity between the precuneus and hippocampus. Enhanced memory performance was linked to increased gamma oscillatory activity and connectivity through white matter tracts.

Strengths:

The combination of personalized repetitive TMS (iTBS) and gamma tACS is a novel approach to targeting the precuneus, and thereby, connected memory-related regions to enhance long-term associative memory. The authors leverage an existing neural mechanism engaged in

memory binding, theta-gamma coupling, by applying TMS at theta burst patterns and tACS at gamma frequencies to enhance gamma oscillations. The authors conducted a thorough study that suggests that simultaneous iTBS and gamma tACS could be a powerful approach for enhancing long-term associative memory. The paper was well-written, clear, and concise.

Comments on Revision:

I thank the authors for their thoughtful responses to my first review and their inclusion of more detailed methodological discussion of their rationale for the stimulation protocol conditions and timing. Regarding the apparent difference in connectivity at baseline between conditions, the explanation that this is due to intrinsic dynamics, state, or noise implies the baseline is reflecting transient changes in dynamics rather than a true or stable baseline. Based on this, it looks like iTBS solely is significantly greater than the baseline before the iTBS and γ tACS condition but maybe not that much lower than post-stimulation period for iTBS and γ tACS. A longer baseline period should be used to ensure transient states are not driving baseline levels such that these endogenous fluctuations would average out. This also raises questions about whether the effect of iTBS and γ tACS or iTBS alone are dependent on the intrinsic state at the time when stimulation begins. Their additional clarification of memory scoring is helpful but also reveals that the effect of dual iTBS+ γ tACS specifically on the association between faces and names is just significant. This modest increase in associative memory should be taken into consideration when interpreting these findings.

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Reviewer #3 (Public review):

Summary:

Borghi and colleagues present results from 4 experiments aimed at investigating the effects of dual γ tACS and iTBS stimulation of the precuneus on behavioral and neural markers of memory formation. In their first experiment ($n = 20$), they find that a 3-minute offline (i.e., prior to task completion) stimulation that combines both techniques leads to superior memory recall performance in an associative memory task immediately after learning associations between pictures of faces, names, and occupation, as well as after a 15-minute delay, compared to iTBS alone (+ tACS sham) or no stimulation (sham for both iTBS and tACS). Performance in a second task probing short-term memory was unaffected by the stimulation condition. In a second experiment ($n = 10$), they show that these effects persist over 24 hours and up to a full week after initial stimulation. A third ($n = 14$) and fourth ($n = 16$) experiment were conducted to investigate neural effects of the stimulation protocol. The authors report that, once again, only combined iTBS and γ tACS increases gamma oscillatory activity and neural excitability (as measured by concurrent TMS-EEG) specific to the stimulated area at the precuneus compared to a control region, as well as precuneus-hippocampus functional connectivity (measured by resting state MRI), which seemed to be associated with structural white matter integrity of the bilateral middle longitudinal fasciculus (measured by DTI).

Strengths:

Combining non-invasive brain stimulation techniques is a novel, potentially very powerful method to maximize the effects of these kinds of interventions that are usually well-tolerated and thus accepted by patients and healthy participants. It is also very impressive that the stimulation-induced improvements in memory performance resulted from a short (3 min) intervention protocol. If the effects reported here turn out to be as clinically meaningful and generalizable across populations as implied, this approach could represent a promising avenue for treatment of impaired memory functions in many conditions.

Methodologically, this study is expertly done! I don't see any serious issues with the technical setup in any of the experiments. It is also very commendable that the authors conceptually replicated the behavioral effects of experiment 1 in experiment 2 and then conducted two additional experiments to probe the neural mechanisms associated with these effects. This certainly increases the value of the study and the confidence in the results considerably.

The authors used a within-subject approach in their experiments, which increases statistical power and allows for stronger inferences about the tested effects. They also used to individualize stimulation locations and intensities, which should further optimize the signal-to-noise ratio.

Weaknesses:

I think one of the major weaknesses of this study is the overall low sample size in all of the experiments (between $n = 10$ and $n = 20$). This is, as I mentioned when discussing the strengths of the study, partly mitigated by the within-subject design and individualized stimulation parameters. The authors mention that they performed a power analysis but this analysis seemed to be based on electrophysiological readouts similar to those obtained in experiment 3. It is thus unclear whether the other experiments were sufficiently powered to reliably detect the behavioral effects of interest. In the revised manuscript, the authors provide post-hoc sensitivity analyses that help contextualize the strength of the findings.

While the authors went to great lengths trying to probe the neural changes likely associated with the memory improvement after stimulation, it is impossible from their data to causally relate the findings from experiments 3 and 4 to the behavioral effects in experiments 1 and 2. This is acknowledged by the authors and there are good methodological reasons for why TMS-EEG and fMRI had to be collected in separate experiments, but readers should keep in mind that this limits inferences about how exactly dual iTBS and γ tACS of the precuneus modulate learning and memory.

<https://doi.org/10.7554/eLife.104220.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The authors claim that they can use a combination of repetitive transcranial magnetic stimulation (intermittent theta burst-iTBS) and transcranial alternating current stimulation (gamma tACS) to cause slight improvements in memory in a face/name/profession task.

Strengths:

The idea of stimulating the human brain non-invasively is very attractive because, if it worked, it could lead to a host of interesting applications. The current study aims to evaluate one such exciting application.

Weaknesses:

(1) It is highly unclear what, if anything, transpires in the brain with non-invasive stimulation. To cite one example of many, a rigorous study in rats and human cadavers, compellingly showed that traditional parameters of transcranial electrical stimulation

lead to no change in brain activity due to the attenuation by the soft tissue and skull (Mihály Vöröslakos et al Nature Communications 2018): <https://www.nature.com/articles/s41467-018-02928-3>. It would be very useful to demonstrate via invasive neurophysiological recordings that the parameters used in the current study do indeed lead to any kind of change in brain activity. Of course, this particular study uses a different non-invasive stimulation protocol.

Thank you for raising the important issue regarding the actual neurophysiological effects of non-invasive brain stimulation. Unfortunately, invasive neurophysiological recordings in humans for this type of study are not feasible due to ethical constraints, while studies on cadavers or rodents would not fully resolve our question. Indeed, the authors of the cited study (Mihály Vöröslakos et al., Nature Communications, 2018) highlight the impossibility of drawing definitive conclusions about the exact voltage required in the in-vivo human brain due to significant differences between rats and humans, as well as the in-vivo human brain and cadavers due to alterations in electrical conductivity that occur in postmortem tissue. Huang and colleagues addressed the difficulties in reaching direct evidence of non-invasive brain stimulation (NIBS) effects in a review published in Clinical Neurophysiology in 2017. They conclude that the use of EEG to assess brain response to TMS has great potential for a less indirect demonstration of plasticity mechanisms induced by NIBS in humans.

To address this challenge, we conducted Experiments 3 and 4, which respectively examined the neurophysiological and connectivity changes induced by the stimulation in a non-invasive manner using TMS-EEG and fMRI. The observed changes in brain oscillatory activity (increased gamma oscillatory activity), cortical excitability (enhanced posteromedial parietal cortex reactivity), and brain connectivity (strengthened connections between the precuneus and hippocampi) provided evidence of the effects of our non-invasive brain stimulation protocol, further supporting the behavioral data.

Additionally, we carefully considered the issue of stimulation distribution and, in response, performed a biophysical modeling analysis and E-field calculation using the parameters employed in our study (see Supplementary Materials).

We acknowledge that further exploration of this aspect would be highly valuable, and we agree that it is worth discussing both as a technical limitation and as a potential direction for future research. We therefore, modify the discussion accordingly (main text, lines 280-289).

“Although we studied TMS and tACS propagation through the E-field modeling and observed an increase in the precuneus gamma oscillatory activity, excitability and connectivity with the hippocampi, we cannot exclude that our results might reflect the consequences of stimulating more superficial parietal regions other than the precuneus nor report direct evidence of microscopic changes in the brain after the stimulation. Invasive neurophysiological recordings in humans for this type of study are not feasible due to ethical constraints. Studies on cadavers or rodents would not fully resolve our question due to significant differences between them (i.e. rodents do not have an anatomical correspondence while cadavers have an alterations in electrical conductivity occurring in postmortem tissue). However, further exploration of this aspect in future studies would help in the understanding of ytACS+iTBS effects.”

(2) If there is any brain activity triggered by the current stimulation parameters, then it is extremely difficult to understand how this activity can lead to enhancing memory. The brain is complex. There are hundreds of neuronal types. Each neuron receives precise input from about 10,000 other neurons with highly tuned synaptic strengths. Let us assume that the current protocol does lead to enhancing (or inhibiting) simultaneously the activity of millions of neurons. It is unclear whether there is any activity at all in the brain triggered by this protocol, it is also unclear whether such activity would be

excitatory, or inhibitory. It is also unclear how many neurons, let alone what types of neurons would change their activity. How is it possible that this can lead to memory enhancement? This seems like using a hammer to knock on my laptop and hope that the laptop will output a new Mozart-like sonata.

Thank you for your comment. As you correctly point out, we still do not have precise knowledge of which neurons—and to what extent—are activated during non-invasive brain stimulation in humans. However, this challenge is not limited to brain stimulation but applies to many other therapeutic interventions, including psychiatric medications, without limiting their use.

Nevertheless, a substantial body of research has investigated the mechanisms underlying the efficacy of TMS and tACS in producing behavioral after-effects, primarily through its ability to induce long-term potentiation (Bliss & Collingridge, The Journal of Physiology, 1993a; Ridding & Rothwell, Nature Reviews Neuroscience, 2007; Huang et al., Clinical Neurophysiology, 2017; Koch et al., Neuroimage 2018; Koch et al., Brain 2022; Jannati et al., Neuropsychopharmacology, 2023; Wischniewski et al., Trends in Cognitive Science, 2023; Griffiths et al., Trends in Neuroscience, 2023).

We acknowledge that we took this important aspect for granted. We consequently expanded the introduction accordingly (main text, lines 48-60).

“Repetitive transcranial magnetic stimulation (rTMS) and transcranial alternating current stimulation (tACS) are two forms of NIBS widely used to enhance memory performances (Grover et al., 2022; Koch et al., 2018; Wang et al., 2014). rTMS, based on the principle of Faraday, induces depolarization of cortical neuronal assemblies and leads to after-effects that have been linked to changes in synaptic plasticity involving mechanisms of long-term potentiation (LTP) (Huang et al., 2017; Jannati et al., 2023). On the other hand, tACS causes rhythmic fluctuations in neuronal membrane potentials, which can bias spike timing, leading to an entrainment of the neural activity (Wischniewski et al., 2023). In particular, the induction of gamma oscillatory activity has been proposed to play an important role in a type of LTP known as spike timing-dependent plasticity, which depends on a precise temporal delay between the firing of a presynaptic and a postsynaptic neuron (Griffiths and Jensen, 2023). Both LTP and gamma oscillations have a strong link with memory processes such as encoding (Bliss and Collingridge, 1993; Griffiths and Jensen, 2023; Rossi et al., 2001), pointing to rTMS and tACS as good candidates for memory enhancement.”

(3) Even if there is any kind of brain activation, it is unclear why the authors seem to be so sure that the precuneus is responsible. Are there neurophysiological data demonstrating that the current protocol only activates neurons in the precuneus? Of note, the non-invasive measurements shown in Figure 3 are very weak (Figure 3A top and bottom look very similar, and Figure 3C left and right look almost identical). Even if one were to accept the weak alleged differences in Figure 3, there is no indication in this figure that there is anything specific to the precuneus, rather a whole brain pattern. This would be the kind of minimally rigorous type of evidence required to make such claims. In a less convincing fashion, one could look at different positions of the stimulation apparatus. This would not be particularly compelling in terms of making a statement about the precuneus. But at least it would show that the position does matter, and over what range of distances it matters, if it matters.

Thank you for your feedback. Our assumption that the precuneus plays a key role in the observed effects is based on several factors:

(1) The non-invasive stimulation protocol was applied to an individually identified precuneus for each participant. Given existing evidence on TMS propagation, we can reasonably assume

that the precuneus was at least a mediator of the observed effects (Ridding & Rothwell, Nature Reviews Neuroscience 2007). For further details about target identification and TMS and tACS propagation, please refer to the MRI data acquisition section in the main text and Biophysical modeling and E-field calculation section in the supplementary materials.

(2) To investigate the effects of the neuromodulation protocol on cortical responses, we conducted a whole-brain analysis using multiple paired t-tests comparing each data point between different experimental conditions. To minimize the type I error rate, data were permuted with the Monte Carlo approach and significant p-values were corrected with the false discovery rate method (see the Methods section for details). The results identified the posterior-medial parietal areas as the only regions showing significant differences across conditions.

(3) To control for potential generalized effects, we included a control condition in which TMS-EEG recordings were performed over the left parietal cortex (adjacent to the precuneus). This condition did not yield any significant results, reinforcing the cortical specificity of the observed effects.

However, as stated in the Discussion, we do not claim that precuneus activity alone accounts for the observed effects. As shown in Experiment 4, stimulation led to connectivity changes between the precuneus and hippocampus, a network widely recognized as a key contributor to long-term memory formation (Bliss & Collingridge, Nature 1993). These connectivity changes suggest that precuneus stimulation triggered a ripple effect extending beyond the stimulation site, engaging the broader precuneus-hippocampus network.

Regarding Figure 3A, it represents the overall expression of oscillatory activity detected by TMS-EEG. Since each frequency band has a different optimal scaling, the figure reflects a graphical compromise. A more detailed representation of the significant results is provided in Figure 3B. The effect sizes for gamma oscillatory activity in the delta T1 and T2 conditions were 0.52 and 0.50, respectively, which correspond to a medium effect based on Cohen's d interpretation.

We add a paragraph in the discussion to improve the clarity of the manuscript regarding this important aspect (lines 193-198).

“Given the existing evidence on TMS propagation and the computation of the Biophysical model with the Efield, we can reasonably assume that the individually identified PC was a mediator of the observed effects (Ridding and Rothwell, 2007). Moreover, we observed specific cortical changes in the posteromedial parietal areas, as evidenced by the whole-brain analysis conducted on TMS-EEG data and the absence of effect on the lateral posterior parietal cortex used as a control condition.”

(4) In the absence of any neurophysiological documentation of a direct impact on the brain, an argument in this type of study is that the behavioral results show that there must be some kind of effect. I agree with this argument. This is also the argument for placebo effects, which can be extremely powerful and useful even if the mechanism is unrelated to what is studied. Then let us dig into the behavioral results.

Hoping to have already addressed your concern regarding the neurophysiological impact of the stimulation on the brain, we would like to emphasize that the behavioral results were obtained controlling for placebo effects. This was achieved by having participants perform the task under different stimulation conditions, including a sham condition.

4a. There does not seem to be any effect on the STMB task, therefore we can ignore this.

4b. The FNAT task is minimally described in the supplementary material. There are no experimental details to understand what was done. What was the size of the images? How long were the images presented for? Were there any repetitions of the images? For how long did the participants study the images? Presumably, all the names and occupations are different? What were the genders of the faces? What is chance level performance? Presumably, the same participant saw different faces across the different stimulation conditions. If not, then there can be memory effects across different conditions that are even more complex to study. If yes, then it would be useful to show that the difficulty is the same across the different stimuli.

We thank you for signaling the lack in the description of FNAT task. We added the information required in the supplementary information (lines 93-101).

“Each picture's face size was 19x15cm. In the learning phase, faces were shown along with names and occupations for 8 seconds each (totaling approximately 2 minutes). During immediate recall, the faces were displayed alone for 8 seconds. In the delayed recall and recognition phase, pictures were presented until the subject provided answers. We used a different set of stimuli for each stimulation condition, resulting in a total of 3 parallel task forms balanced across conditions and session order. All parallel forms comprised 6 male and 6 female faces; for each sex, there were 2 young adults (around 30 years old), 2 middle-aged adults (around 50 years old), and 2 elderly adults (around 70 years old). Before the experiments, we conducted a pilot study to ensure no differences existed between the parallel forms of the task.”

The chance level in the immediate and delayed recall is not quantifiable since the participants had to freely recall the name and the occupation without a multiple choice. In the recognition, the chance level was around 33% (since the possible answers were 3).

4c. Although not stated clearly, if I understand FNAT correctly, the task is based on just 12 presentations. Each point in Figure 2A represents a different participant. Unfortunately, there is no way of linking the performance of individual participants across the conditions with the information provided. Lines joining performance for each participant would be useful in this regard. Because there are only 12 faces, the results are quantized in multiples of 100/12 % in Figure 3A. While I do not doubt that the authors did their homework in terms of the statistical analyses, it is difficult to get too excited about these 12 measurements. For example, take Figure 3A immediate condition TOTAL, arguably the largest effect in the whole paper. It seems that on average, the participants may remember one more face/name/occupation.

Thank you for the suggestion. We added graphs showing lines linking the performance of individual participants across conditions to improve clarity, please see Fig.2 revised. We apologize for the lack of clarity in the description of the FNAT. As you correctly pointed out, we used the percentage based on the single association between face, name and occupation (12 in total). However, each association consisted of three items, resulting in a total of 36 items to learn and associate – we added a paragraph to make it more explicit in the manuscript (lines 425-430).

“We considered a correct association when a subject was able to recall all the information for each item (i.e. face, name and occupation), resulting in a total of 36 items to learn and associate. To further investigate the effect on FNAT we also computed a partial recall score accounting for those items where subjects correctly matched only names with faces (FNAT NAME) and only occupations with faces (FNAT OCCUPATION). See supplementary information for score details.”

In the example you mentioned, participants were, on average, able to correctly recall and associate three more items compared to the other conditions. While this difference may not seem striking at first glance, it is important to consider that we assessed memory performance after a single, three-minute stimulation session. Similar effects are typically observed only after multiple stimulation sessions (Koch et al., *NeuroImage*, 2018; Grover et al., *Nature Neuroscience*, 2022). Moreover, memory performance changes are often measured by a limited set of stimuli due to methodological constraints related to memory capacity. For example, Rey Auditory Verbal learning task, requiring to learn and recall 15 words, is a typical test used to detect memory changes (Koch et al., *Neuroimage*, 2018; Benussi et al., *Brain stimulation* 2021; Benussi et al., *Annals of Neurology*, 2022).

4d. Block effects. If I understand correctly, the experiments were conducted in blocks. This is always problematic. Here is one example study that articulated the big problems in block designs (Li et al TPAMI 2021):<https://ieeexplore.ieee.org/document/9264220>

Thank you for the interesting reference. According to this paper, in a block design, EEG or fMRI recordings are performed in response to different stimuli of a given class presented in succession. If this is the case, it does not correspond to our experimental design where both TMS-EEG and fMRI were conducted in resting state on different days according to the different stimulation conditions.

4e. Even if we ignore the lack of experimental descriptions, problems with lack of evidence of brain activity, the minimalistic study of 12 faces, problems with the block design, etc. at the end of the day, the results are extremely weak. In FNAT, some results are statistically significant, some are not. The interpretation of all of this is extremely complex. Continuing with Figure 3A, it seems that the author claims that iTBS+gtACS > iTBS+sham-tACS, but iTBS+gtACS ~ sham+sham. I am struggling to interpret such a result. When separating results by name and occupation, the results are even more perplexing. There is only one condition that is statistically significant in Figure 3A NAME and none in the occupation condition.

Thank you again for your feedback. Hoping to have thoroughly addressed your initial concerns in our previous responses, we now move on to your observations regarding the behavioral results, assuming you were referring to Figure 2A. The main finding of this study is the improvement in long-term memory performance, specifically the ability to correctly recall the association between face, name, and occupation (total FNAT), which was significantly enhanced in both Experiments 1 and 2. However, we also aimed to explore the individual contributions of name and occupation separately to gain a deeper understanding of the results. Our analysis revealed that the improvement in total FNAT was primarily driven by an increase in name recall rather than occupation recall. We understand that this may have caused some confusion. We consequently modified the manuscript in the (lines 97-99; 107-111; 425-430) to make it clearer and moved the graph relative to FNAT NAME and OCCUPATION from fig.2 in the main text to fig. S4 in supplementary information.

“Dual iTBS+ytACS increased the performances in recalling the association between face, name and occupation (FNAT accuracy) both for the immediate ($F_{2,38}=7.18$; $p=0.002$; $\eta^2_p=0.274$) and the delayed ($F_{2,38}=5.86$; $p=0.006$; $\eta^2_p=0.236$) recall performances (Fig. 2, panel A).”

“The in-depth analysis of the FNAT accuracy investigating the specific contribution of face-name and face-occupation recall revealed that dual iTBS+ytACS increased the performances in the association between face and name (FNAT NAME) delayed recall ($F_{2,38}=3.46$; $p=0.042$; $\eta^2_p=0.154$; iTBS+ytACS vs. sham-iTBS+sham-tACS: $42.9\pm 21.5\%$ vs. $33.8\pm 19\%$; $p=0.048$ Bonferroni corrected) (Fig. S4, supplementary information).”

“We considered a correct association when a subject was able to recall all the information for each item (i.e. face, name and occupation), resulting in a total of 36 items to learn and associate. To further investigate the effect on FNAT we also computed a partial recall score accounting for those items where subjects correctly matched only names with faces (FNAT NAME) and only occupations with faces (FNAT OCCUPATION). See supplementary information for score details.”

Regarding the stimulation conditions, your concerns about the performance pattern (iTBS+gtACS > iTBS+sham-tACS, but iTBS+gtACS ~ sham+sham) are understandable. However, this new protocol was developed precisely in response to the variability observed in behavioral outcomes following non-invasive brain stimulation, particularly when used to modulate memory functions (Corp et al., 2020; Pabst et al., 2022). As discussed in the manuscript, it is intended as a boost to conventional non-invasive brain stimulation protocols, leveraging the mechanisms outlined in the Discussion section.

(5) In sum, it would be amazing to be able to use non-invasive stimulation for any kind of therapeutic purpose as the authors imagine. More work needs to be done to convince ourselves that this kind of approach is viable. The evidence provided in this study is weak.

We hope our response will be carefully considered, fostering a constructive exchange and leading to a reassessment of your evaluation.

Reviewer #2 (Public review):

Summary:

The manuscript "Dual transcranial electromagnetic stimulation of the precuneus-hippocampus network boosts human long-term memory" by Borghi and colleagues provides evidence that the combination of intermittent theta burst TMS stimulation and gamma transcranial alternating current stimulation (ytACS) targeting the precuneus increases long-term associative memory in healthy subjects compared to iTBS alone and sham conditions. Using a rich dataset of TMS-EEG and resting-state functional connectivity (rs-FC) maps and structural MRI data, the authors also provide evidence that dual stimulation increased gamma oscillations and functional connectivity between the precuneus and hippocampus. Enhanced memory performance was linked to increased gamma oscillatory activity and connectivity through white matter tracts.

Strengths:

The combination of personalized repetitive TMS (iTBS) and gamma tACS is a novel approach to targeting the precuneus, and thereby, connected memory-related regions to enhance long-term associative memory. The authors leverage an existing neural mechanism engaged in memory binding, theta-gamma coupling, by applying TMS at theta burst patterns and tACS at gamma frequencies to enhance gamma oscillations. The authors conducted a thorough study that suggests that simultaneous iTBS and gamma tACS could be a powerful approach for enhancing long-term associative memory. The paper was well-written, clear, and concise.

Weaknesses:

(1) The study did not include a condition where ytACS was applied alone. This was likely because a previous work indicated that a single 3-minute ytACS did not produce significant effects, but this limits the ability to isolate the specific contribution of ytACS in the context of this target and memory function

Thank you for your comments. As you pointed out, we did not include a condition where ytACS was applied alone. This decision was based on the findings of Guerra et al. (Brain Stimulation 2018), who investigated the same protocol and reported no aftereffects. Given the substantial burden of the experimental design on patients and our primary goal of demonstrating an enhancement of effects compared to the standalone iTBS protocol, we decided to leave out this condition. However, you raise an important aspect that should be further discussed, we modified the limitation section accordingly (lines 290-297).

“We did not assess the effects of ytACS alone. This decision was based on the findings of Guerra et al. (Guerra et al., 2018), who investigated the same protocol and reported no aftereffects. Given the substantial burden of the experimental design on patients and our primary goal of demonstrating an enhancement of effects compared to the standalone iTBS protocol, we decided to leave out this condition. While examining the effects of ytACS alone could help isolate its specific contribution to this target and memory function, extensive research has shown that achieving a cognitive enhancement aftereffect with tACS alone typically requires around 20–25 minutes of stimulation (Grover et al., 2023).”

(2) The authors applied stimulation for 3 minutes, which seems to be based on prior tACS protocols. It would be helpful to present some rationale for both the duration and timing relative to the learning phase of the memory task. Would you expect additional stimulation prior to recall to benefit long-term associative memory?

Thank you for your comment and for raising this interesting point. As you correctly noted, the protocol we used has a duration of three minutes, a choice based on previous studies demonstrating its greater efficacy with respect to single stimulation from a neurophysiological point of view. Specifically, these studies have shown that the combined stimulation enhanced gamma-band oscillations and increased cortical plasticity (Guerra et al., Brain Stimulation 2018; Maiella et al., Scientific Reports 2022). Given that the precuneus (Brodt et al., Science 2018; Schott et al., Human Brain Mapping 2018), gamma oscillations (Osipova et al., Journal of Neuroscience 2006; Deprés et al., Neurobiology of Aging 2017; Griffiths et al., Trends in Neurosciences 2023), and cortical plasticity (Brodt et al., Science 2018) are all associated with memory formation and encoding processes, we decided to apply the co-stimulation immediately before it to enhance the efficacy. We added this paragraph to the manuscript rationale (lines 48-60).

“Repetitive transcranial magnetic stimulation (rTMS) and transcranial alternating current stimulation (tACS) are two forms of NIBS widely used to enhance memory performances (Grover et al., 2022; Koch et al., 2018; Wang et al., 2014). rTMS, based on the principle of Faraday, induces depolarization of cortical neuronal assemblies and leads to after-effects that have been linked to changes in synaptic plasticity involving mechanisms of long-term potentiation (LTP) (Huang et al., 2017; Jannati et al., 2023). On the other hand, tACS causes rhythmic fluctuations in neuronal membrane potentials, which can bias spike timing, leading to an entrainment of the neural activity (Wischnewski et al., 2023). In particular, the induction of gamma oscillatory a has been proposed to play an important role in a type of LTP known as spike timing-dependent plasticity, which depends on a precise temporal delay between the firing of a presynaptic and a postsynaptic neuron (Griffiths and Jensen, 2023). Both LTP and gamma oscillations have a strong link with memory processes such as encoding (Bliss and Collingridge, 1993; Griffiths and Jensen, 2023; Rossi et al., 2001), pointing to rTMS and tACS as good candidates for memory enhancement.”

Regarding the question of whether stimulation could also benefit recall, the answer is yes. We can speculate that repeating the stimulation before recall might provide an additional boost. This is supported by evidence showing that both the precuneus and gamma oscillations are involved in recall processes (Flanagin et al., Cerebral Cortex 2023; Griffiths et al., Trends in

Neurosciences 2023). Furthermore, previous research suggests that reinstating the same brain state as during encoding can enhance recall performance (Javadi et al., The Journal of Neuroscience 2017). We added this consideration to the discussion (lines 305-311).

“Future studies should further investigate the effects of stimulation on distinct memory processes. In particular, stimulation could be applied before retrieval (Rossi et al., 2001), to better elucidate its specific contribution to the observed enhancements in memory performance. Additionally, it would be worth examining whether repeated stimulation - administered both before encoding and before retrieval - could produce a boosting effect. This is especially relevant in light of findings showing that matching the brain state between retrieval and encoding can significantly enhance memory performance (Javadi et al., 2017).”

(3) How was the burst frequency of theta iTBS and gamma frequency of tACS chosen? Were these also personalized to subjects' endogenous theta and gamma oscillations? If not, were increases in gamma oscillations specific to patients' endogenous gamma oscillation frequencies or the tACS frequency?

The stimulation protocol was chosen based on previous studies (Guerra et al., Brain Stimulation 2018; Maiella et al., Scientific Reports 2022). Gamma tACS sinusoid frequency wave was set at 70 Hz while iTBS consisted of ten bursts of three pulses at 50 Hz lasting 2 s, repeated every 10 s with an 8 s pause between consecutive trains, for a total of 600 pulses total lasting 190 s (see iTBS+γtACS neuromodulation protocol section). In particular, the theta iTBS has been inspired by protocols used in animal models to elicit LTP in the hippocampus (Huang et al., Neuron 2005). Consequently, neither Theta iTBS nor the gamma frequency of tACS were personalized. The increase in gamma oscillations was referred to the patient's baseline and did not correspond to the administered tACS frequency.

(4) The authors do a thorough job of analyzing the increase in gamma oscillations in the precuneus through TMS-EEG; however, the authors may also analyze whether theta oscillations were also enhanced through this protocol due to the iTBS potentially targeting theta oscillations. This may also be more robust than gamma oscillations increases since gamma oscillations detected on the scalp are very low amplitude and susceptible to noise and may reflect activity from multiple overlapping sources, making precise localization difficult without advanced techniques.

Thank you for the suggestion. We analyzed theta oscillations, finding no changes.

(5) Figure 4: Why are connectivity values pre-stimulation for the iTBS and sham tACS stimulation condition so much higher than the dual stimulation? We would expect baseline values to be more similar.

We acknowledge that the pre-stimulation connectivity values for the iTBS and sham tACS conditions appear higher than those for the dual stimulation condition. However, as noted in our statistical analyses, there were no significant differences at baseline between conditions ($p\text{-FDR}=0.3514$), suggesting that any apparent discrepancy is due to natural variability rather than systematic bias. One potential explanation for these differences is individual variability in baseline connectivity measures, which can fluctuate due to factors such as intrinsic neural dynamics, participant state, or measurement noise. Despite these variations, our statistical approach ensures that any observed post-stimulation effects are not confounded by pre-existing differences.

(6) Figure 2: How are total association scores significantly different between stimulation conditions, but individual name and occupation associations are not? Further clarification of how the total FNAT score is calculated would be helpful.

We apologize for any lack of clarity. The total FNAT score reflects the ability to correctly recall all the information associated with a person—specifically, the correct pairing of the face, name, and occupation. Participants received one point for each triplet they accurately recalled. The scores were then converted into percentages, as detailed in the Face-Name Associative Task Construction and Scoring section in the supplementary materials.

Total FNAT was the primary outcome measure. However, we also analyzed name and occupation recall separately to better understand their partial contributions. Our analysis revealed that the improvement in total FNAT was primarily driven by an increase in name recall rather than occupation recall.

We acknowledge that this distinction may have caused some confusion. To improve clarity, we revised the manuscript accordingly (lines 97-98; 107-111; 425-430).

“Dual iTBS+ytACS increased the performances in recalling the association between face, name and occupation (FNAT accuracy) both for the immediate ($F_{2,38}=7.18$; $p=0.002$; $\eta^2_p=0.274$) and the delayed ($F_{2,38}=5.86$; $p=0.006$; $\eta^2_p=0.236$) recall performances (Fig. 2, panel A).”

“The in-depth analysis of the FNAT accuracy investigating the specific contribution of face-name and face-occupation recall revealed that dual iTBS+ytACS increased the performances in the association between face and name (FNAT NAME) delayed recall ($F_{2,38}=3.46$; $p=0.042$; $\eta^2_p=0.154$; iTBS+ytACS vs. sham-iTBS+sham-tACS: 42.9 ± 21.5 % vs. 33.8 ± 19 %; $p=0.048$ Bonferroni corrected) (Fig. S4, supplementary information).”

“We considered a correct association when a subject was able to recall all the information for each item (i.e. face, name and occupation), resulting in a total of 36 items to learn and associate. To further investigate the effect on FNAT we also computed a partial recall score accounting for those items where subjects correctly matched only names with faces (FNAT NAME) and only occupations with faces (FNAT OCCUPATION). See supplementary information for score details.”

We also moved the data regarding the specific contribution of name and occupation recall in the supplementary information (fig.S4) and further specified how we computed the score in the score (lines 102-104).

“The score was computed by deriving an accuracy percentage index dividing by 12 and multiplying by 100 the correct association sum. The partial recall scores were computed in the same way only considering the sum of face-name (NAME) and face-occupation (OCCUPATION) correctly recollected.”

Reviewer #3 (Public review):

Summary:

Borghi and colleagues present results from 4 experiments aimed at investigating the effects of dual ytACS and iTBS stimulation of the precuneus on behavioral and neural markers of memory formation. In their first experiment (n = 20), they found that a 3-minute offline (i.e., prior to task completion) stimulation that combines both techniques leads to superior memory recall performance in an associative memory task immediately after learning associations between pictures of faces, names, and occupation, as well as after a 15-minute delay, compared to iTBS alone (+ tACS sham) or no stimulation (sham for both iTBS and tACS). Performance in a second task probing short-term memory was unaffected by the stimulation condition. In a second experiment (n = 10), they show that these effects persist over 24 hours and up to a full week after initial stimulation. A third (n = 14) and fourth (n = 16) experiment were conducted to investigate the neural effects of the stimulation protocol. The authors report that, once again, only combined iTBS and

ytACS increase gamma oscillatory activity and neural excitability (as measured by concurrent TMS-EEG) specific to the stimulated area at the precuneus compared to a control region, as well as precuneus-hippocampus functional connectivity (measured by resting-state MRI), which seemed to be associated with structural white matter integrity of the bilateral middle longitudinal fasciculus (measured by DTI).

Strengths:

Combining non-invasive brain stimulation techniques is a novel, potentially very powerful method to maximize the effects of these kinds of interventions that are usually well-tolerated and thus accepted by patients and healthy participants. It is also very impressive that the stimulation-induced improvements in memory performance resulted from a short (3 min) intervention protocol. If the effects reported here turn out to be as clinically meaningful and generalizable across populations as implied, this approach could represent a promising avenue for the treatment of impaired memory functions in many conditions.

Methodologically, this study is expertly done! I don't see any serious issues with the technical setup in any of the experiments (with the only caveat that I am not an expert in fMRI functional connectivity measures and DTI). It is also very commendable that the authors conceptually replicated the behavioral effects of experiment 1 in experiment 2 and then conducted two additional experiments to probe the neural mechanisms associated with these effects. This certainly increases the value of the study and the confidence in the results considerably.

The authors used a within-subject approach in their experiments, which increases statistical power and allows for stronger inferences about the tested effects. They are also used to individualize stimulation locations and intensities, which should further optimize the signal-to-noise ratio.

Weaknesses:

I want to state clearly that I think the strengths of this study far outweigh the concerns I have. I still list some points that I think should be clarified by the authors or taken into account by readers when interpreting the presented findings.

I think one of the major weaknesses of this study is the overall low sample size in all of the experiments (between $n = 10$ and $n = 20$). This is, as I mentioned when discussing the strengths of the study, partly mitigated by the within-subject design and individualized stimulation parameters. The authors mention that they performed a power analysis but this analysis seemed to be based on electrophysiological readouts similar to those obtained in experiment 3. It is thus unclear whether the other experiments were sufficiently powered to reliably detect the behavioral effects of interest. That being said, the authors do report significant effects, so they were per definition powered to find those. However, the effect sizes reported for their main findings are all relatively large and it is known that significant findings from small samples may represent inflated effect sizes, which may hamper the generalizability of the current results. Ideally, the authors would replicate their main findings in a larger sample. Alternatively, I think running a sensitivity analysis to estimate the smallest effect the authors could have detected with a power of 80% could be very informative for readers to contextualize the findings. At the very least, however, I think it would be necessary to address this point as a potential limitation in the discussion of the paper.

Thank you for the observation. As you mentioned, our power analysis was based on our previous study investigating the same neuromodulation protocol with a corresponding experimental design. The relatively small sample could be considered a possible limitation of

the study which we will add to the discussion. A fundamental future step will be to replay these results on a larger population, however, to strengthen our results we performed the sensitivity analysis you suggested.

In detail, we performed a sensitivity analysis for repeated-measures ANOVA with $\alpha=0.05$ and power $(1-\beta)=0.80$ with no sphericity correction. For experiment 1, a sensitivity analysis with 1 group and 3 measurements showed a minimal detectable effect size of $f=0.524$ with 20 participants. In our paper, the ANOVA on total FNAT immediate performance revealed an effect size of $\eta^2=0.274$ corresponding to $f=0.614$; the ANOVA on FNAT delayed performance revealed an effect size of $\eta^2=0.236$ corresponding to $f=0.556$. For experiment 2, a sensitivity analysis for total FNAT immediate performance (1 group and 3 measurements) showed a minimal detectable effect size of $f=0.797$ with 10 participants. In our paper, the ANOVA on total FNAT immediate performance revealed an effect size of $\eta^2=0.448$ corresponding to $f=0.901$. The sensitivity analysis for total FNAT delayed performance (1 group and 6 measurements) showed a minimal detectable effect size of $f=0.378$ with 10 participants. In our paper, the ANOVA on total FNAT delayed performance revealed an effect size of $\eta^2=0.484$ corresponding to $f=0.968$. Thus, the sensitivity analysis showed that both experiments were powered enough to detect the minimum effect size computed in the power analysis. We have now added this information to the manuscript and we thank the reviewer for her/his suggestion in the statistical analysis and results section (lines 99-100; 127-128; 130-131; 543-545).

“The sensitivity analysis showed a minimal detectable effect size of $\eta^2=0.215$ with 20 participants.”

“The sensitivity analysis showed a minimal detectable effect size of $\eta^2=0.388$ with 10 participants.”

“The sensitivity analysis showed a minimal detectable effect size of $\eta^2=0.125$ with 10 participants.”

“Since we do not have an a priori effect size for experiment 1 and 2, we performed a sensitivity power analysis to ensure that these experiments were able to detect the minimum effect size with 80% power and alpha level of 0.05.”

It seems that the statistical analysis approach differed slightly between studies. In experiment 1, the authors followed up significant effects of their ANOVAs by Bonferroni-adjusted post-hoc tests whereas it seems that in experiment 2, those post-hoc tests were "exploratory", which may suggest those were uncorrected. In experiment 3, the authors use one-tailed t-tests to follow up their ANOVAs. Given some of the reported p-values, these choices suggest that some of the comparisons might have failed to reach significance if properly corrected. This is not a critical issue per se, as the important test in all these cases is the initial ANOVA but non-significant (corrected) post-hoc tests might be another indicator of an underpowered experiment. My assumptions here might be wrong, but even then, I would ask the authors to be more transparent about the reasons for their choices or provide additional justification. Finally, the authors sometimes report exact p-values whereas other times they simply say $p < .05$. I would ask them to be consistent and recommend using exact p-values for every result where $p \geq .001$.

Thank you again for the suggestions. Your observations are correct, we used a slightly different statistical depending on our hypothesis. Here are the details:

In experiment 1, we used a repeated-measure ANOVA with one factor “stimulation condition” (iTBS+ytACS; iTBS+sham-tACS; sham-iTBS+sham-tACS). Following the significant effect of this factor we performed post-hoc analysis with Bonferroni correction.

In experiment 2, we used a repeated-measures with two factors “stimulation condition” and “time”. As expected, we observed a significant effect of condition, confirming the result of experiment 1, but not of time. Thus, this means that the neuromodulatory effect was present regardless of the time point. However, to explore whether the effects of stimulation condition were present in each time point we performed some explorative t-tests with no correction for multiple comparisons since this was just an explorative analysis.

In experiment 3, we used the same approach as experiment 1. However, since we had a specific hypothesis on the direction of the effect already observed in our previous study, i.e. increase in spectral power (Maiella et al., Scientific Report 2022), our tests were 1-tailed.

For the p-values, we corrected the manuscript reporting the exact values for every result.

While the authors went to great lengths trying to probe the neural changes likely associated with the memory improvement after stimulation, it is impossible from their data to causally relate the findings from experiments 3 and 4 to the behavioral effects in experiments 1 and 2. This is acknowledged by the authors and there are good methodological reasons for why TMS-EEG and fMRI had to be collected in sperate experiments, but it is still worth pointing out to readers that this limits inferences about how exactly dual iTBS and ytACS of the precuneus modulate learning and memory.

Thank you for your comment. We fully agree with your observation, which is why this aspect has been considered in the study's limitations. To address your concern, we add this sentence to the limitation discussion (lines 299-301).

“Consequently, these findings do not allow precise inferences regarding the specific mechanisms by which dual iTBS and ytACS of the precuneus modulate learning and memory.”

There were no stimulation-related performance differences in the short-term memory task used in experiments 1 and 2. The authors argue that this demonstrates that the intervention specifically targeted long-term associative memory formation. While this is certainly possible, the STM task was a spatial memory task, whereas the LTM task relied (primarily) on verbal material. It is thus also possible that the stimulation effects were specific to a stimulus domain instead of memory type. In other words, could it be possible that the stimulation might have affected STM performance if the task taxed verbal STM instead? This is of course impossible to know without an additional experiment, but the authors could mention this possibility when discussing their findings regarding the lack of change in the STM task.

Thank you for your interesting observation. We argue that the intervention primarily targeted long-term associative memory formation, as our findings demonstrated effects only on FNAT. However, as you correctly pointed out, we cannot exclude the possibility that the stimulation may also influence short-term verbal associative memory. We add this aspect when discussing the absence of significant findings in the STM task (lines 205-210).

“Visual short-term associative memory, measured by STBM performance, was not modulated by any experimental condition. Even if we cannot exclude the possibility that the stimulation could have influenced short-term verbal associative memory, we expected this result since short-term associative memory is known to rely on a distinct frontoparietal network while FNAT, used to investigate long-term associative memory, has already been associated with the neural activity of the PC and the hippocampus (Parra et al., 2014; Rentz et al., 2011).”

While the authors discuss the potential neural mechanisms by which the combined stimulation conditions might have helped memory formation, the psychological

processes are somewhat neglected. For example, do the authors think the stimulation primarily improves the encoding of new information or does it also improve consolidation processes? Interestingly, the beneficial effect of dual iTBS and ytACS on recall performance was very stable across all time points tested in experiments 1 and 2, as was the performance in the other conditions. Do the authors have any explanation as to why there seems to be no further forgetting of information over time in either condition when even at immediate recall, accuracy is below 50%? Further, participants started learning the associations of the FNAT immediately after the stimulation protocol was administered. What would happen if learning started with a delay? In other words, do the authors think there is an ideal time window post-stimulation in which memory formation is enhanced? If so, this might limit the usability of this procedure in real-life applications.

Thank you for your comment and for raising these important points.

We hypothesized that co-stimulation would enhance encoding processes. Previous studies have shown that co-stimulation can enhance gamma-band oscillations and increase cortical plasticity (Guerra et al., Brain Stimulation 2018; Maiella et al., Scientific Reports 2022). Given that the precuneus (Brodt et al., Science 2018; Schott et al., Human Brain Mapping 2018), gamma oscillations (Osipova et al., Journal of Neuroscience 2006; Deprés et al., Neurobiology of Aging 2017; Griffiths et al., Trends in Neurosciences 2023), and cortical plasticity (Brodt et al., Science 2018) have all been associated with encoding processes, we decided to apply co-stimulation before the encoding phase, to boost it. We enlarged the introduction to specify the link between neural mechanisms and the psychological process of the encoding (lines 55-60).

“In particular, the induction of gamma oscillatory activity has been proposed to play an important role in a type of LTP known as spike timing-dependent plasticity, which depends on a precise temporal delay between the firing of a presynaptic and a postsynaptic neuron (Griffiths and Jensen, 2023). Both LTP and gamma oscillations have a strong link with memory processes such as encoding (Bliss and Collingridge, 1993; Griffiths and Jensen, 2023; Rossi et al., 2001), pointing to rTMS and tACS as good candidates for memory enhancement.”

We applied the co-stimulation immediately before the learning phase to maximize its potential effects. While we observed a significant increase in gamma oscillatory activity lasting up to 20 minutes, we cannot determine whether the behavioral effects we observed would have been the same with a co-stimulation applied 20 minutes before learning. Based on existing literature, a reduction in the efficacy of co-stimulation over time could be expected (Huang et al., Neuron 2005; Thut et al., Brain Topography 2009). However, we hypothesize that multiple stimulation sessions might provide an additional boost, helping to sustain the effects over time (Thut et al., Brain Topography 2009; Koch et al., Neuroimage 2018; Koch et al., Brain 2022).

Regarding the absence of further forgetting in both stimulation conditions, we think that the clinical and demographical characteristics of the sample (i.e. young and healthy subjects) explain the almost absence of forgetting after one week.

Reviewer #1 (Recommendations for the authors):

To address the concerns, the authors should:

(1) Include invasive neuronal recordings (e.g., in rats or monkeys if not possible in humans) demonstrating that the current stimulation protocol leads to direct changes in brain activity.

We understand the interest of the first reviewer in the understanding of neurophysiological correlates of the stimulation protocol, however, we are skeptical about this request as we

think it goes beyond the aims of the study. As already mentioned in the response to the reviewer, invasive neurophysiological recordings in humans for this type of study are not feasible due to ethical constraints. At the same time, studies on cadavers or rodents would not fully resolve the question. Indeed, the authors of the study cited by the reviewer (Mihály Vöröslakos et al., Nature Communications, 2018) highlight the impossibility of drawing definitive conclusions about the exact voltage required in the in-vivo human brain due to significant differences between rats and humans, as well as the in-vivo human cadavers due to alterations in electrical conductivity that occur in postmortem tissue. Huang and colleagues addressed the difficulties in reaching direct evidence of non-invasive brain stimulation (NIBS) effects in a review published in Clinical Neurophysiology in 2017. They conclude that the use of EEG to assess brain response to TMS has a great potential for a less indirect demonstration of plasticity mechanisms induced by NIBS in humans.

It is exactly to meet the need to investigate the changes in brain activity after the stimulation protocol that we conducted Experiments 3 and 4. These experiments respectively examined the neurophysiological and connectivity changes induced by the stimulation in a non-invasive manner using TMS-EEG and fMRI. The observed changes in brain oscillatory activity (increased gamma oscillatory activity), cortical excitability (enhanced posteromedial parietal cortex reactivity), and brain connectivity (strengthened connections between the precuneus and hippocampi) provided evidence of the effects of our non-invasive brain stimulation protocol, further supporting the behavioral data.

Additionally, we carefully considered the issue of stimulation distribution and, in response, performed a biophysical modeling analysis and E-field calculation using the parameters employed in our study (see Supplementary Materials).

Acknowledging the reviewer's point of view, we modified the manuscript accordingly, discussing this aspect both as a technical limitation and as a potential direction for future research (main text, lines 280-289).

“Although we studied TMS and tACS propagation through the E-field modeling and observed an increase in the precuneus gamma oscillatory activity, excitability and connectivity with the hippocampi, we cannot exclude that our results might reflect the consequences of stimulating more superficial parietal regions other than the precuneus nor report direct evidence of microscopic changes in the brain after the stimulation. Invasive neurophysiological recordings in humans for this type of study are not feasible due to ethical constraints. Studies on cadavers or rodents would not fully resolve our question due to significant differences between them (i.e. rodents do not have an anatomical correspondence while cadavers have an alterations in electrical conductivity occurring in postmortem tissue). However, further exploration of this aspect in future studies would help in the understanding of γ tACS+iTBS effects.”

| (2) Address all the technical questions about the experimental design.

We addressed all the technical questions about the experimental design.

| (3) Repeat the experiments with randomized trial order and without a block design.

The experiments were conducted with randomized trial order and we did not use a block design.

| (4) Add many more faces to the study. It is extremely difficult to draw any conclusion from merely 12 faces. Ideally, there would be lots of other relevant memory experiments where the authors show compelling positive results.

We understand your perplexity about drawing conclusions from 12 faces, however, this is not the case. As we explained in the response reviewer, the task we implemented did not rely on the recall of merely 12 faces. Instead, participants had to correctly learn, associate and recall 12 faces, 12 names and 12 occupations for a total of 36 items. To improve the clarity of the manuscript, we added a paragraph to make this aspect more explicit (lines 425-430).

“We considered a correct association when a subject was able to recall all the information for each item (i.e. face, name and occupation), resulting in a total of 36 items to learn and associate. To further investigate the effect on FNAT we also computed a partial recall score accounting for those items where subjects correctly matched only names with faces (FNAT NAME) and only occupations with faces (FNAT OCCUPATION). See supplementary information for score details.”

The behavioral changes we observed are similar to those who are typically observed after multiple stimulation sessions (Koch et al., *NeuroImage*, 2018; Grover et al., *Nature Neuroscience*, 2022, Benussi et al., *Annals of Neurology*, 2022). Moreover, memory performance changes are often measured by a limited set of stimuli due to methodological constraints related to memory capacity. For example, Rey Auditory Verbal learning task, requiring to learn and recall 15 words, is a typical test used to detect memory changes (Koch et al., *Neuroimage*, 2018; Benussi et al., *Brain stimulation* 2021; Benussi et al., *Annals of Neurology*, 2022).

(5) Provide a clear explanation of the apparent randomness of which results are statistically significant or not in Figure 3. But perhaps with many more experiments, a lot more memory evaluations, many more stimuli, and addressing all the other technical concerns, either the results will disappear or there will be a more interpretable pattern of results.

We provided explanations for all the concerns shown by the reviewer.

Reviewer #2 (Recommendations for the authors):

Minor comments:

(1) Figure 4: Why are connectivity values pre-stimulation for the iTBS and sham tACS stimulation condition so much higher than the dual stimulation? We would expect baseline values to be more similar.

We acknowledge that the pre-stimulation connectivity values for the iTBS and sham tACS conditions appear higher than those for the dual stimulation condition. However, as noted in our statistical analyses, there were no significant differences at baseline between conditions ($p\text{-FDR} = 0.3514$), suggesting that any apparent discrepancy is due to natural variability rather than systematic bias. One potential explanation for these differences is individual variability in baseline connectivity measures, which can fluctuate due to factors such as intrinsic neural dynamics, participant state, or measurement noise. Despite these variations, our statistical approach ensures that any observed post-stimulation effects are not confounded by pre-existing differences.

(2) Figure 2: How are total association scores significantly different between stimulation conditions, but individual name and occupation associations are not? Further clarification of how the total FNAT score is calculated would be helpful.

We apologize for any lack of clarity. The total FNAT score reflects the ability to correctly recall all the information associated with a person—specifically, the correct pairing of the face, name, and occupation. Participants received one point for each triplet they accurately

recalled. The scores were then converted into percentages, as detailed in the Face-Name Associative Task Construction and Scoring section in the supplementary materials.

Total FNAT was the primary outcome measure. However, we also analyzed name and occupation recall separately to better understand their partial contributions. Our analysis revealed that the improvement in total FNAT was primarily driven by an increase in name recall rather than occupation recall.

We acknowledge that this distinction may have caused some confusion. To improve clarity, we revised the manuscript accordingly (lines 97-98; 107-111; 425-430).

“Dual iTBS+ytACS increased the performances in recalling the association between face, name and occupation (FNAT accuracy) both for the immediate ($F_{2,38}=7.18$; $p=0.002$; $\eta^2_p=0.274$) and the delayed ($F_{2,38}=5.86$; $p=0.006$; $\eta^2_p=0.236$) recall performances (Fig. 2, panel A).”

“The in-depth analysis of the FNAT accuracy investigating the specific contribution of face-name and face-occupation recall revealed that dual iTBS+ytACS increased the performances in the association between face and name (FNAT NAME) delayed recall ($F_{2,38}=3.46$; $p=0.042$; $\eta^2_p=0.154$; iTBS+ytACS vs. sham-iTBS+sham-tACS: $42.9\pm 21.5\%$ vs. $33.8\pm 19\%$; $p=0.048$ Bonferroni corrected) (Fig. S4, supplementary information).”

“We considered a correct association when a subject was able to recall all the information for each item (i.e. face, name and occupation), resulting in a total of 36 items to learn and associate. To further investigate the effect on FNAT we also computed a partial recall score accounting for those items where subjects correctly matched only names with faces (FNAT NAME) and only occupations with faces (FNAT OCCUPATION). See supplementary information for score details.”

We also moved the data regarding the specific contribution of name and occupation recall in the supplementary information (fig.S4) and further specified how we computed the score in the score (lines 102-104).

“The score was computed by deriving an accuracy percentage index dividing by 12 and multiplying by 100 the correct association sum. The partial recall scores were computed in the same way only considering the sum of face-name (NAME) and face-occupation (OCCUPATION) correctly recollected.”

Reviewer #3 (Recommendations for the authors):

A very small detail, in the caption for Figure 2A, OCCUPATION is described as being shown on the 'left' but it should be 'right'.

We corrected this error.

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